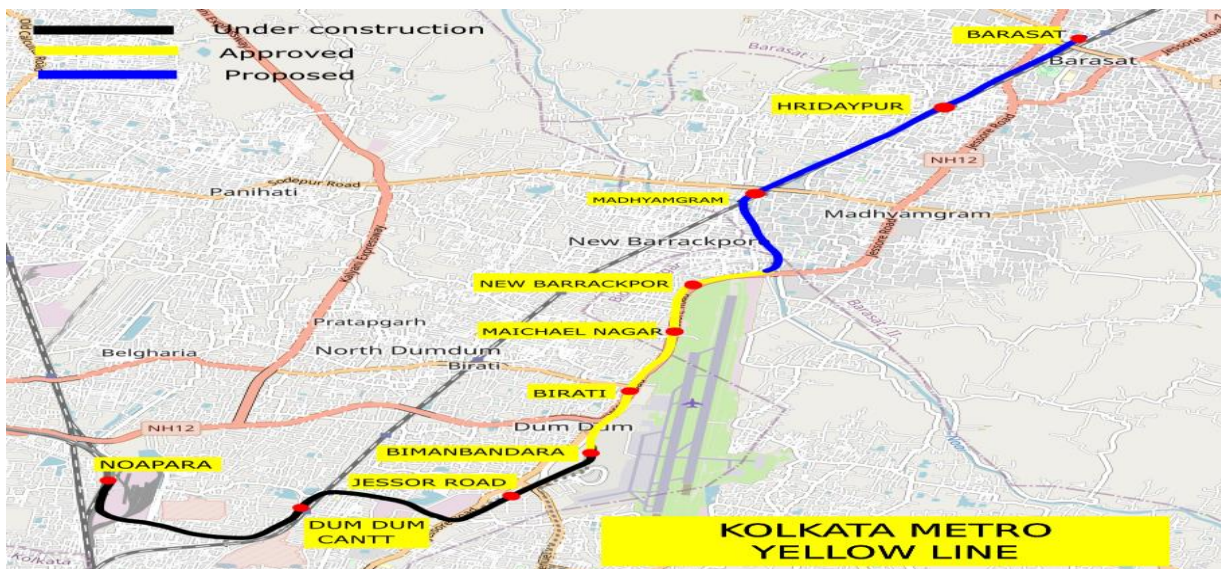




**PROJECT REPORT ON INDUSTRIAL TRAINING  
AT  
BIRATI UG METRO, METRO RAILWAY, KOLKATA**

**SUBMITTED BY THE STUDENTS OF  
“Indian Institute of Engineering Science and Technology, Shibpur”**





**INDIAN INSTITUTE OF ENGINEERING SCIENCE AND TECHNOLOGY,  
SHIBPUR**

Howrah-711103 ,West Bengal,India

**Submitted By**

**Name : SOUPARNA SANNYASI**

**Roll No : 2023CEB025**

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## ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Cemindia Projects Limited** for providing me with the opportunity to undergo my industrial training at the **Birati Underground Metro Project, Metro Railway, Kolkata**. The training provided me with valuable practical exposure to various aspects of underground metro construction and helped me understand how the concepts learned in the classroom are applied in real engineering projects.

I am sincerely thankful to **Mr. Goutam Sikdar**, Project Manager, along with the management, engineers, supervisors and all the staff members of **Cemindia Projects Limited** for their invaluable guidance, encouragement, support and cooperation throughout the training period. Their professional insights, practical knowledge and willingness to share their experience greatly enhanced my understanding of modern construction practices and project execution, making this training a truly valuable learning experience.

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Finally, I would like to thank my parents and friends for their constant support and encouragement throughout the training period and during the preparation of this report.

The knowledge and experience gained during this training have enhanced my understanding of Civil Engineering practices and will be beneficial in my academic as well as professional career.

SOUPARNA SANNYASI

Department of Civil Engineering

IEST, Shibpur

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**SIGNATURE OF THE STUDENT**

## **Introduction to Cemindia Projects Limited**

Cemindia Projects Limited is one of India's leading engineering and construction companies, engaged in the execution of large-scale infrastructure and heavy civil engineering projects. The company traces its origins to 1931 and was incorporated as an Indian entity in 1978. Formerly known as ITD Cementation India Limited, it was renamed Cemindia Projects Limited in 2025, continuing its long-standing legacy of delivering technically complex infrastructure projects.

Headquartered in Mumbai, the company operates as an Engineering, Procurement and Construction (EPC) organization, providing end-to-end solutions for a wide range of infrastructure sectors. Over the years, it has successfully executed numerous landmark projects for both government and private sector clients, contributing significantly to India's infrastructure development.

The major areas of operation of Cemindia Projects Limited include:

- Metro Rail and Mass Rapid Transit Systems (MRTS)
- Tunnels and Underground Structures
- Highways, Bridges and Flyovers
- Marine Structures, Ports, Jetties and Harbours
- Airports and Industrial Infrastructure
- Dams, Irrigation Projects and Hydraulic Structures
- Water Supply and Wastewater Infrastructure
- Foundation Engineering and Specialized Civil Works

The company is widely recognized for its expertise in executing technically demanding projects using modern construction technologies, advanced equipment, and stringent quality control procedures. It places strong emphasis on workplace safety, environmental responsibility, timely project delivery and adherence to national and international engineering standards.

The industrial training for this report was carried out at the Birati Underground Metro Project, Metro Railway, Kolkata, one of the company's major ongoing metro infrastructure projects. During the training, I had the opportunity to observe the construction practices adopted by the company including diaphragm wall construction, guide wall construction, reinforcement works, concrete batching, quality control procedures and safety management practices. The exposure provided valuable practical insight into the planning and execution of large-scale underground metro construction projects.

## **Introduction to Birati UG Metro Project, Metro Railway, Kolkata**

The Birati Underground Metro Project is a significant part of the expansion of the Kolkata Metro network, aimed at improving urban transportation and connectivity in the northern suburban region of Kolkata. The project is located at Birati, a rapidly developing area near Netaji Subhas Chandra Bose International Airport.

The project is being executed by Cemindia Projects Limited (formerly ITD Cementation India Limited). It is a part of the Yellow Line (Line-4) extension, connecting the airport (Jai Hind Bimanbandar Station) to Birati with the proposed extension further towards Barasat.

Design, Procurement & Construction of RCC Box Metro tunnel including Birati UG Station building & Subways (with Entry-Exit Structures) for two underground Metro Stations (Birati & Michael Nagar) by Cut & Cover Method (Top Down with Diaphragm Wall/Bottom Up Method with support arrangement including architectural finishing, firefighting, water supply, sanitary installation, drainage & BL,1 works, F&M service, ECS Service, TVS Services, Power Supply System with 750 VDC Traction System including composite Aluminium third Rail, from Ch: 6.250 km to Ch: 9.840 km including all ancillary works in connection with extension of Metro Railway from Noapara to Barasat via Biman Bandar.

The scope of the Birati UG Metro Project includes the construction of:

- A 3.5–3.6 km underground metro tunnel
- A fully underground metro station at Birati
- Subways and entry/exit structures at Birati and Michael Nagar
- Installation of Mechanical, Electrical and Plumbing (MEP) systems, track work, tunnel ventilation, fire safety, and drainage systems.

The construction involves advanced engineering techniques such as:

- Cut-and-cover method for underground station construction
- RCC box pushing method for construction near the airport boundary
- Use of diaphragm walls (D-wall) to ensure soil stability and safety during deep excavation.

This project is of great importance because it:

- Enhances connectivity between Kolkata city and its northern suburbs
- Reduces traffic congestion and travel time
- Provides direct metro connectivity to the airport
- Promotes sustainable and efficient urban transportation

The Birati Underground Metro Project is one of the major infrastructure developments in Kolkata and demonstrates the application of modern construction techniques, advanced engineering practices and effective project management in the execution of underground metro systems.

## Project Details

**Project Title:** Kolkata Metro Line-4 (Yellow Line) Underground Extension – Biman Bandar (Airport) to Birati

**Client:** Metro Railway Kolkata

**Executing Contractor:** ITD Cementation India Ltd.

**Contract Type:** Engineering, Procurement and Construction (EPC)

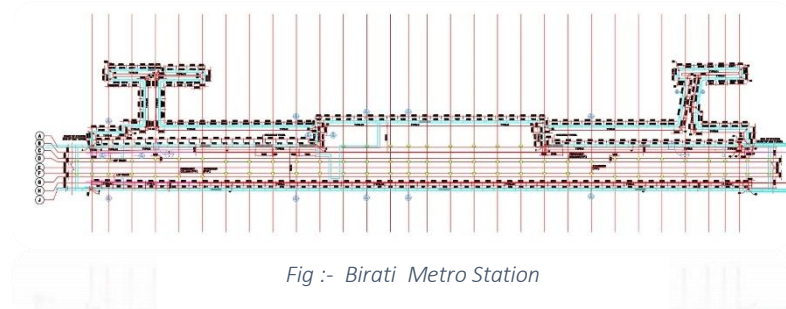
**Contract Duration:** 910 Days (Approx. 2.5 Years)

**Contract Value:** ₹1,097.23 Crore

**Tender No.** EPC-2489-2024

The Kolkata Metro Line-4 (Yellow Line) Underground Extension Project is a key urban transportation project undertaken by Metro Railway Kolkata to improve connectivity between the airport and the northern suburbs of Kolkata. The internship was carried out on the Biman Bandar (Airport)–Birati underground section, which forms part of the 11.26 km Noapara–Barasat Metro Corridor. The project is being executed by ITD Cementation India Ltd. under an EPC (Engineering, Procurement and Construction) contract.

The scope of this package includes the construction of approximately 3.59 km of underground metro tunnel, Birati Underground Metro Station, passenger subways at Birati and Michael Nagar, and all associated civil, structural, architectural and MEP works. The tunnel along the airport boundary is being constructed using the RCC Box Pushing Method, while Birati Station is being built using the Cut-and-Cover Method with diaphragm walls for excavation support and groundwater control. In addition to civil works, the project includes installation of ballastless tracks, 750 V DC Third Rail traction system, electrical systems, tunnel ventilation, firefighting systems, water supply, drainage and architectural finishes. With a contract value of approximately ₹1,097 crore and a construction period of 910 days, the project is expected to significantly improve public transportation, reduce traffic congestion and enhance connectivity in North Kolkata. The project also provides valuable exposure to modern underground metro construction techniques, quality control, safety practices and multidisciplinary project execution.

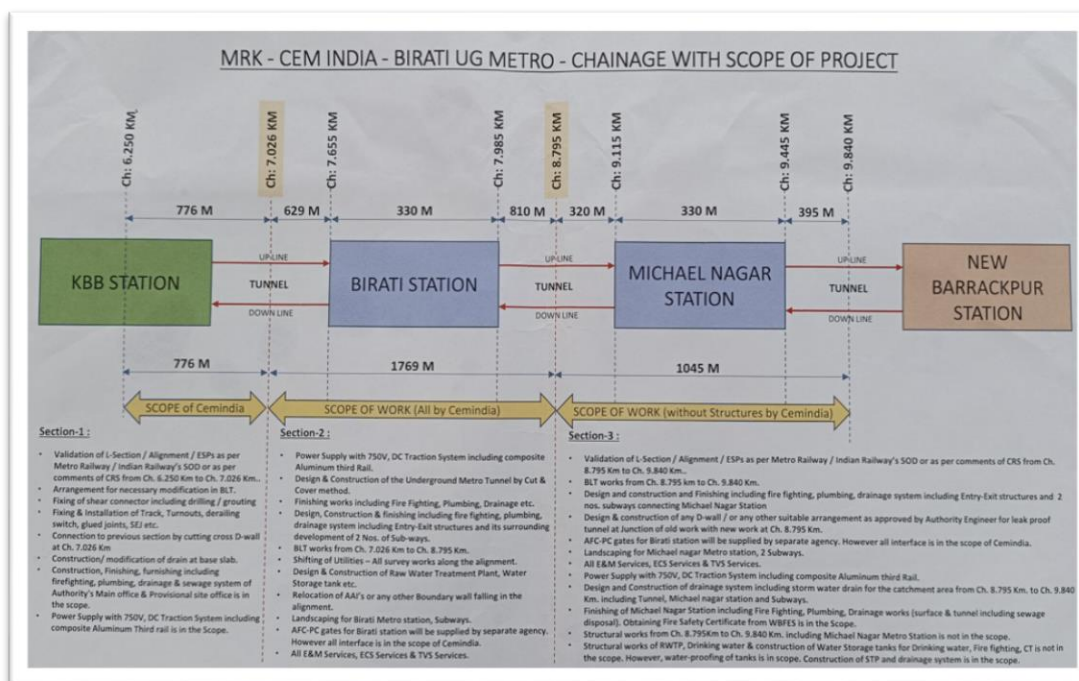


During the initial stage of construction, existing Traffic Guard building was there within the proposed alignment of the project. To facilitate construction, the structure was demolished after obtaining the necessary approvals and following all prescribed safety procedures. After demolition, the construction alignment was adjusted by providing an appropriate offset to accommodate site constraints and optimize the execution of the works.

## Scope of Work

- Construction of Underground Metro Corridor:
  - Construction of approximately 3.59 km of underground metro tunnel between Biman Bandar (Airport) and Birati as part of the Noapara–Barasat Metro Corridor.
- Construction of Birati Underground Station:
  - Development of the underground metro station, including station box, platforms, concourse level, service rooms and passenger circulation areas using the Cut-and-Cover method.
- Construction of Passenger Subways:
  - Construction of pedestrian subways with entry and exit structures at Birati and Michael Nagar stations to provide safe passenger access.
- Tunnel Construction:
  - Construction of reinforced concrete metro tunnels using the RCC Box Pushing Method along the airport boundary to minimize disturbance to airport operations.
- Diaphragm Wall and Deep Excavation Works:
  - Execution of diaphragm walls for earth retention and groundwater control, followed by deep excavation for station and tunnel construction.
- Structural Concrete Works:
  - Construction of foundation slabs, base slabs, retaining walls, columns, beams, roof slabs and other reinforced concrete structural components.
- Architectural Finishing Works:
  - Installation of flooring, wall cladding, ceilings, painting, doors, windows, staircases and other architectural finishes to make the station operational.
- Mechanical, Electrical and Plumbing (MEP) Works:
  - Installation of water supply, drainage, sanitary systems, firefighting systems, electrical distribution, lighting, tunnel ventilation (TVS) and Environmental Control System (ECS).
- Track and Traction System Installation:
  - Laying of ballastless metro tracks and installation of the 750 V DC Third Rail traction system along with associated electrical infrastructure.
- Utility Shifting and Site Development:
  - Relocation of existing underground and overhead utilities, traffic management, road restoration and site development to facilitate uninterrupted construction.
- Quality Assurance and Safety Management:
  - Implementation of quality control procedures through material testing, inspections and adherence to safety standards, including PPE usage, toolbox talks and safe work practices.
- Testing and Commissioning:

- Inspection, testing and commissioning of civil structures, electrical systems, mechanical systems and metro infrastructure before handing over the project for commercial operation.



**Pictorial representation of scope of project**

### List of Major Construction Activities:

1. Survey & Geo-Technical Investigation work
2. Development of Design-Drawings & Obtaining Approval from Employer/Authority Engineer.
3. Identification & Relocation/Shifting of Utilities including Boundary Walls & Storm-Water drain falling in the alignment from Ch 7.026 Km to Ch 9.840 Km.
4. Construction & Delivery of Project-Facilities of Employer/Authority Engineer.
5. Diaphragm Wall Construction.
6. Piling Work (Pile & Plunge Pile Construction).
7. Excavation Work
8. Strut-Waler Structural Fabrication & Erection Work above Roof Slab.
9. Construction of Roof Slab of RCC Box Structure.
10. Waterproofing above Roof Slab.

11. Construction of Concourse Slab with OTE duct for Birati Station Portion.
12. Construction of Base Slab including Waterproofing below Base Slab.
13. Construction of Platform Slabs with UPE ducts, Columns, Beams & Walls.
14. Construction of Staircase, Lift, Escalators & Ancillary Building Structures with Chiller Plants & Cooling Towers.
15. Construction of Subways & Entry-Exit Structures-02 Nos each for both Birati & Michael Nagar Stations.
16. Architectural Finishing including Fire-fighting, Plumbing, drainage & Landscaping work for Birati & Michael Nagar stations including Subways & Entry-Exit Structures.
17. Ballast-Less Track-Work from Biman Bandar Yard (Ch 7.026 Km) to Buffer end of Michael Nagar Station (Ch 9.840 Km) including necessary modification of Existing BLT work from Ch: 6.250 Km to Ch: 7.026 Km.
18. Construction of Raw-Water Treatment Plant, Sewage Treatment Plant. Water Storage Tanks etc. for both Birati & Michael Nagar stations.
19. Construction of Road Services including Backfilling over Cut & Cover & Stations including Railing. Drain, Footpath & Boundary Wall as per Scope & Specification.
20. Electrical work for Manufacture, Delivery, Installation & Testing & Commissioning of EBM, ECS & TVS Services.
21. Electric Power Supply with 750V DC Traction System including Composite Aluminum Third Rail.

## Major Construction Methods

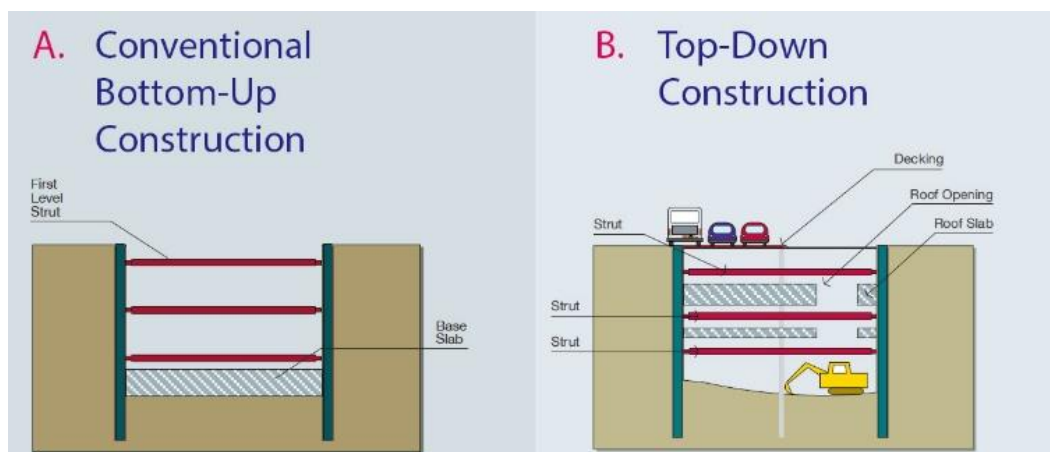
### 1. Diaphragm Wall Construction:

Diaphragm walls are constructed using a Grabber Machine and Polymer slurry to form reinforced concrete retaining walls. These walls provide earth retention, groundwater control and permanent structural support during deep excavation.



### 2. Cut-and-Cover Method:

This method is used for constructing Birati Underground Station. After constructing the diaphragm walls, the soil is excavated, followed by the construction of the base slab, side walls, roof slab and backfilling. It is suitable for shallow underground stations.



### **3. RCC Box Pushing Method:**

The underground tunnel along the airport boundary is constructed by hydraulically pushing reinforced concrete box segments through the soil. This method eliminates the need for Tunnel Boring Machines (TBMs) and minimizes disruption to airport operations and surrounding infrastructure.

#### **Objectives of the Project**

- Improve metro connectivity between the airport, Birati and the Barasat corridor.
- Reduce travel time and road congestion in North Kolkata.
- Provide safe, efficient and sustainable urban transportation.
- Develop modern underground metro infrastructure with integrated civil and systems works.

## **Safety Induction Programme**

On the first day of my internship at Cemindia Projects Limited (formerly ITD ,Cementation India Limited), we attended a mandatory Safety Induction Programme before entering the construction site. The programme was conducted to familiarize interns with the basic safety rules, mandatory Personal Protective Equipment (PPE), potential hazards associated with construction activities and the safety procedures to be followed while working at the project site.

During the induction, we were informed that construction sites contain numerous hazards arising from heavy machinery, lifting operations, excavation works, reinforcement activities and movement of construction vehicles. Therefore, strict compliance with the safety regulations and the proper use of PPE are essential to minimize occupational risks.

#### **Personal Protective Equipment (PPE)**

One of the major topics covered during our induction was the proper use of Personal Protective Equipment (PPE). We were informed that PPE acts as the last line of defence against workplace hazards and is mandatory for every person entering the construction site.

##### **Safety helmet:**

We were informed that the safety helmet protects workers against injuries caused by falling tools, construction materials and accidental impacts with structural members.

Since overhead activities are continuously performed at metro construction projects, helmets are mandatory throughout the working.



**Safety Equipments used**

### **Safety Shoes**

Safety shoes with reinforced steel toes protect workers from injuries caused by falling objects, protruding reinforcement bars, sharp construction materials and uneven walking surfaces. They also provide improved grip while moving across muddy or irregular ground.

### **Reflective Safety Vest**

The reflective safety vest increases worker visibility which enables crane operators, equipment operators and construction vehicle drivers to identify personnel from a safe distance, particularly during low-light conditions.

We were also informed that additional PPE such as gloves, safety goggles, ear protection, respiratory protection and full-body safety harnesses are used depending upon the nature of work being carried out.

### Activity-wise recommended PPE

| SL. NO | Activity                                | Suggested PPE                                                                               |
|--------|-----------------------------------------|---------------------------------------------------------------------------------------------|
| 1.     | General site movement                   | Safety helmet, reflective safety vest, safety shoes                                         |
| 2.     | Handling diesel and lubricants          | Rubber hand gloves, safety goggles, safety shoes                                            |
| 3.     | Excavation work                         | Safety helmet, reflective safety vest, safety shoes                                         |
| 4.     | Reinforcement fixing                    | Safety helmet, cut-resistant gloves, safety shoes                                           |
| 5.     | Concreting and batching operations      | Safety helmet, gloves, safety goggles, safety shoes                                         |
| 6.     | Cutting, grinding and welding           | Face shield/safety goggles, leather gloves, safety helmet, safety shoes, hearing protection |
| 7.     | Working at height                       | Full-body safety harness, safety helmet with chin strap, safety shoes                       |
| 8.     | Lifting operations / Crane working area | Safety helmet, reflective safety vest, safety shoes                                         |
| 9.     | Electrical maintenance                  | Insulated gloves, insulated safety shoes, safety helmet                                     |
| 10.    | Dust-generating activities              | Dust mask/respirator, safety goggles, safety helmet, safety shoes                           |

## **Safe Movement Within the Construction Site**

During the induction, we were advised to remain alert while moving around the construction site. Since heavy lifting operations are carried out continuously, particular attention should be given while passing near cranes. Before crossing the crane operating area, we were instructed to ensure that the crane operator had noticed our movement so that lifting operations could be safely controlled.

We were also instructed to remain cautious near excavation pits and floor openings as these areas present significant fall hazards. The instructor emphasized that even seemingly minor negligence can result in serious accidents.

Another important instruction was to avoid unnecessary shortcuts while moving around the project site. Although shortcuts may appear to save time, they often require crossing active work zones or unsafe areas. Following designated walkways and maintaining situational awareness were strongly emphasized throughout the induction.

## **Occupational Health & Safety Management System of Cemindia Projects Limited:**

In addition to the practical safety induction attended at the project site, I studied the Occupational Health & Safety (OH&S) Management System adopted by Cemindia Projects Limited through the company's official safety publications and management documents. The study helped me understand that safety management within the organization is based on a structured management system rather than individual safety practices.

The company follows an Integrated Management System (IMS) that integrates Quality Management (ISO 9001:2015), Environmental Management (ISO 14001:2015) and Occupational Health & Safety Management (ISO 45001:2018). This integrated approach enables quality, environmental protection and worker safety to be considered simultaneously throughout the execution of construction projects.

## **Occupational Safety & Risk Assessment**

One of the key components of the company's OH&S Management System is Procedure P22 – Occupational Safety & Risk Assessment. Under this procedure, hazards associated with both routine and non-routine construction activities are identified before work begins. The risks to personnel, equipment, materials and property are then assessed and appropriate control measures are implemented to reduce the possibility of accidents.

Similarly, the company also follows Procedure P3 – Environmental Aspects through which environmental impacts arising from construction activities are identified and managed as part of the Integrated Management System.

The emphasis on hazard identification before execution reflects the company's proactive approach towards accident prevention rather than relying solely on corrective measures after an incident.

### **Occupational Safety & Risk Assessment**

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The emphasis on hazard identification before execution reflects the company's proactive approach towards accident prevention rather than relying solely on corrective measures after an incident.

### **Job Safety Assessment (JSA)**

For activities involving higher levels of risk, the company performs Job Safety Assessments (JSA) before commencement of work. The assessment identifies potential hazards associated with the activity and determines the necessary control measures, safe work procedures and Personal Protective Equipment (PPE) required for execution.

### **Toolbox Talks and Safety Awareness:**

The company places significant emphasis on creating safety awareness among employees and contract workers through regular communication and training.

One of the most important initiatives is the Daily Toolbox Talks (TBTs) before the commencement of work. These discussions brief workers on the hazards associated with the day's activities, mandatory PPE requirements and important safety precautions.

Apart from Toolbox Talks, the company also conducts:

- Weekly safety reviews.
- Monthly safety awareness programmes.
- Quarterly Corporate EHS reviews.
- Periodic safety campaigns and worker participation activities.

These programmes help reinforce safe working practices and encourage workers to actively participate in maintaining a safe working environment.

### **Stop Work Programme (SHOP 26)**

A noteworthy feature of the company's Occupational Health & Safety Management System is SHOP 26 – Stop Work Programme (SWP).

This programme empowers every employee and worker with both the authority and responsibility to stop any activity that appears unsafe or poses an immediate danger to personnel, equipment or the environment.

The activity is resumed only after the identified hazard has been reviewed and suitable corrective measures have been implemented. This initiative reflects the company's philosophy that safety should never be compromised for productivity.

### **Safety Inspections, Audits and Monitoring**

To ensure effective implementation of the Occupational Health & Safety Management System, the company follows a structured inspection and audit programme.

The system includes:

- Regular project-level safety inspections.
- Corporate EHS inspections.
- Internal management audits.
- External audits conducted for maintenance of the Integrated Management System certifications.

Inspection findings are analysed, corrective measures are implemented and compliance is monitored to ensure continual improvement of safety performance.

### **Incident Reporting and Continuous Improvement**

The company's safety management system emphasizes learning from incidents and near misses.

Whenever required, the organization undertakes:

- Review of unsafe acts.
- Review of unsafe conditions.
- Preparation of safety alerts.
- Revision of risk assessments.

- Implementation of corrective and preventive actions.
  - Communication of lessons learnt across project sites.
- This process ensures that experiences gained from one project contribute towards improving safety performance throughout the organization.

### **Emergency Preparedness**

Emergency preparedness forms an essential part of the company's Occupational Health & Safety Management System.

Project sites maintain emergency response arrangements, first-aid facilities and trained personnel capable of providing immediate assistance during emergencies. Workers are encouraged to report injuries and unsafe conditions promptly so that appropriate action can be taken without delay.

The company's management system also emphasizes preparedness through training, awareness and clearly defined emergency response procedures.

### **Safety Awareness Publications**

Another notable initiative of the company is the publication of *Safe Sites*, a periodic safety awareness newsletter that promotes best practices in construction safety.

The publication covers topics such as:

- Personal Protective Equipment (PPE)
  - Working at height
  - Scaffolding safety
  - Housekeeping
  - Fire prevention
  - Electrical safety
  - Lifting operations
  - Excavation safety
  - Environmental protection
  - Health awareness
  - Safety campaigns and employee participation
- These publications help in spreading practical safety knowledge throughout the organization and reinforce the company's commitment to developing a strong safety culture.

## Overall Understanding

From both the site induction and the company's documents, I understood that construction safety is achieved through a combination of engineering controls, management procedures, worker participation and continuous improvement. Rather than depending on a single safety measure, the company integrates hazard identification, risk assessment, inspections, training, audits and worker involvement into a comprehensive management framework. This systematic approach plays a significant role in ensuring the safe execution of large-scale infrastructure projects while protecting employees, contractors and visitors.

## Management System Certifications

During my visit to the project office, I observed several management system certifications displayed prominently. These certifications reflect Cemindia Projects Limited's commitment to maintaining internationally recognized standards in quality management, environmental protection and occupational health & safety.

- *ISO 9001:2015 – Quality Management System:*

This certification demonstrates that the company follows a structured Quality Management System to ensure consistency in planning, execution, monitoring and continuous improvement of construction activities.

- *ISO 14001:2015 – Environmental Management System:*

This certification signifies the company's commitment towards environmental protection by minimizing pollution, managing waste efficiently, conserving natural resources and complying with applicable environmental regulations during construction activities.

- *ISO 45001:2018 – Occupational Health & Safety Management System:*

This certification demonstrates that the company has established a systematic Occupational Health & Safety Management System to identify workplace hazards, assess associated risks and implement suitable preventive measures. It also promotes worker participation and compliance with the mandatory safety requirements.

**Integrated Management System (IMS):** Together, *ISO 9001:2015*, *ISO 14001:2015* and *ISO 45001:2018* form the company's Integrated Management System (IMS). The integration of these standards enables the organization to simultaneously maintain construction quality, environmental sustainability and workplace safety throughout project execution.



FIG : ISO 9001:2015, ISO 14001:2015 and ISO 45001:2018 Certifications Displayed  
at the Cemindia Projects Limited Site Office

## STRUCTURAL OVERVIEW OF BIRATI UNDERGROUND METRO STATION AND TUNNEL PROJECT

**A Detailed Study of Underground Metro Station Construction Using Cut-and-Cover and Tunnel Boring Machine (TBM) Techniques**

**Constructed by :** Cemindia Construction Pvt. Ltd.

**Project Owner :** Metro Railway, Kolkata

**Design Consultant :** SYSTRA

### INTRODUCTION

#### Introduction

Rapid urbanization and the continuous increase in population have significantly increased the demand for efficient, safe, and sustainable public transportation systems in metropolitan cities. Kolkata, being one of the oldest metropolitan cities in India, experiences severe traffic congestion due to increasing vehicular movement, narrow roads, and dense commercial as well as residential developments. To overcome these transportation challenges, Metro Railway Kolkata has continuously expanded its metro network by constructing new corridors using both elevated and underground alignments.

The Birati Underground Metro Station is a part of the Kolkata Metro expansion project under the Noapara–Barasat Metro Corridor (Yellow Line). The project has been developed to provide a faster, safer, and environmentally friendly transportation system connecting northern Kolkata with the airport region and the rapidly developing suburban areas of Barasat. Since this corridor passes through densely populated urban locations, several sections have been designed as underground structures to minimize land acquisition, reduce traffic disturbances, and avoid interference with existing infrastructure.

The Birati Underground Metro Station is one of the major underground stations along this corridor. The construction of this station involved advanced civil engineering techniques including deep excavation, temporary earth retaining systems, reinforced concrete underground structures, waterproofing systems, underground tunnel construction, utility diversion, and installation of modern railway infrastructure. The project required careful coordination among structural engineers, geotechnical engineers, survey engineers, architects, mechanical engineers, electrical engineers, planners, and construction management teams to ensure safe and efficient execution.

The structural construction work of the Birati Underground Metro Station and its associated tunnel sections was executed by **Cemindia Construction Pvt. Ltd.** The project was designed under the guidance of **SYSTRA**, the engineering consultant responsible for preparing structural drawings, construction methodology, and technical supervision. During the site visit, various stages of construction were observed, including tunnel construction, sheet pile

installation, diaphragm wall construction, temporary support systems, waterproofing, raft foundation, reinforced concrete tunnel box construction, and preparation for track installation.

The Birati underground section incorporates two major underground construction techniques depending on the surrounding site conditions. In comparatively open areas, the **Cut-and-Cover Method** was adopted, whereas in densely populated regions having limited working space and sensitive structures, the **Tunnel Boring Machine (TBM) Method** was employed. The selection of the construction methodology was primarily governed by soil conditions, groundwater level, available construction space, traffic movement, and environmental considerations.

One of the most important aspects of underground metro construction is maintaining stability of the surrounding soil during excavation. Large excavation depths generate significant lateral earth pressure, groundwater pressure, and surcharge loads from nearby roads and buildings. Therefore, temporary retaining structures such as steel sheet piles, diaphragm walls, steel walers, and steel struts were installed to support the excavation before the permanent reinforced concrete tunnel structure was completed. Once the reinforced concrete walls and roof slabs attained sufficient strength, these temporary support systems were gradually removed, allowing the permanent tunnel box to safely resist all structural loads.

Apart from structural construction, the project also involved extensive preliminary activities before excavation could begin. Detailed topographical surveys, utility mapping, geotechnical investigations, traffic studies, and structural condition assessments of nearby buildings were carried out to minimize construction risks. Borehole investigations were conducted at regular intervals to determine the soil profile, groundwater conditions, bearing capacity, and engineering properties of the subsurface soil. Based on these investigations, the final tunnel alignment, excavation methodology, retaining structures, and foundation systems were designed to ensure structural safety and long-term performance.

Construction quality control played an equally important role throughout the project. Continuous monitoring of excavation depth, lateral wall movement, settlement of adjacent buildings, groundwater level, and vibration generated during sheet pile installation was carried out using modern instrumentation. Since the project is located close to the airport region and other important urban infrastructure, strict monitoring was essential to ensure that construction activities remained within permissible safety limits. During sheet pile driving, crawler cranes equipped with vibratory hammers were used to install Z-type steel sheet piles. Because vibration generated during this operation could potentially affect nearby structures and underground utilities, vibration monitoring was conducted by certified agencies and verified by the concerned government authorities.

The Birati Underground Metro Project also demonstrates the integration of structural engineering with architectural, mechanical, electrical, plumbing (MEP), fire protection, signaling, communication, ventilation, and railway systems. After completion of the civil structures, the underground station accommodates railway tracks, electrical power supply,

tunnel ventilation systems, drainage systems, firefighting installations, passenger platforms, lifts, escalators, communication systems, signaling equipment, and architectural finishes. The successful integration of these multidisciplinary systems transforms the underground concrete structure into a fully operational metro station capable of safely transporting thousands of passengers every day.

The site visit to the Birati Underground Metro Station provided valuable practical exposure to modern underground construction technology. It offered an opportunity to understand how theoretical concepts studied in structural engineering, geotechnical engineering, construction engineering, and surveying are implemented in real-life infrastructure projects. The observations made during the visit significantly enhanced understanding of deep excavation techniques, temporary earth-retaining systems, tunnel construction methodologies, structural sequencing, waterproofing, quality control procedures, and construction safety practices. The experience demonstrated the complexity of underground metro construction and highlighted the importance of careful planning, precise surveying, advanced engineering design, and systematic execution for the successful completion of large-scale urban infrastructure projects.

### **Overview of the Birati Underground Metro Project**

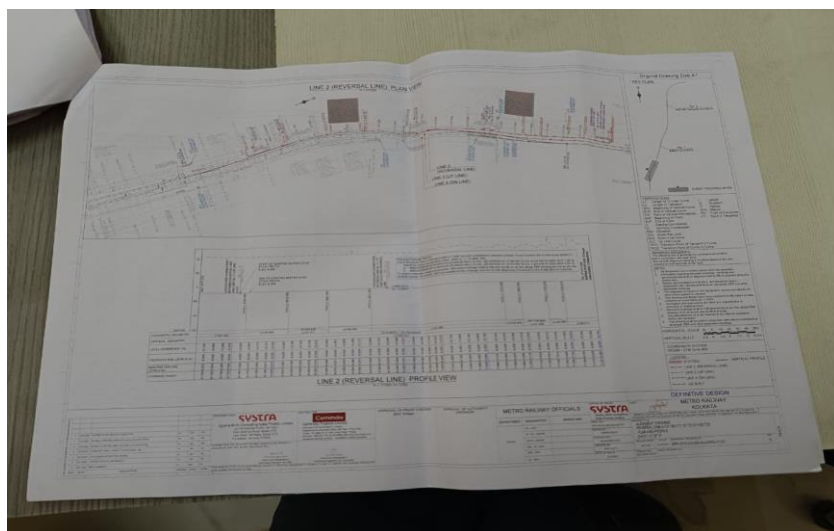
The Birati Underground Metro Station forms an integral part of the Kolkata Metro Railway's Yellow Line (Noapara–Barasat Corridor), which has been developed to improve the transportation network in the northern part of Kolkata. The corridor is designed to connect important residential, commercial, educational, and transportation hubs, thereby reducing travel time, minimizing traffic congestion, and promoting sustainable urban mobility. Since this alignment passes through highly populated urban areas where land acquisition is difficult and traffic movement is extremely heavy, several portions of the metro corridor have been designed as underground structures.

The Birati Underground Metro Station has been constructed using modern underground construction techniques to ensure safe and uninterrupted transportation while minimizing disturbance to existing roads, nearby buildings, underground utilities, and public services. The construction of an underground metro station is considerably more complex than an elevated station because it involves deep excavation, groundwater control, temporary retaining systems, structural waterproofing, reinforced concrete tunnel construction, utility shifting, and continuous monitoring of surrounding structures throughout the construction period.

The civil construction works for the Birati Underground Metro Station were executed by **Cemindia Construction Pvt. Ltd.**, a reputed infrastructure construction company with experience in large-scale transportation projects. The structural design, engineering drawings, construction methodology, and technical supervision were carried out by **SYSTRA**, an internationally recognized engineering consultancy specializing in metro and railway infrastructure.

According to the project alignment observed during the site visit, the Birati Underground Metro Station lies between Jai Hind (Airport) Station and Michael Nagar Station. The underground tunnel section extends between Birati and Michael Nagar, forming a critical link within the Noapara–Barasat Metro Corridor. During the site visit, the construction team explained that different tunnel construction methods were adopted depending on the surrounding site conditions. The **Cut-and-Cover Method** was used in comparatively open stretches where sufficient working space was available, while the **Tunnel Boring Machine (TBM) Method** was employed in highly congested urban areas where surface excavation was not feasible. This selection of construction methodology was based on engineering considerations such as soil profile, groundwater conditions, traffic density, availability of working space, and the presence of nearby buildings and utilities.

The Birati Underground Metro Project demonstrates the integration of advanced structural engineering, geotechnical engineering, surveying, construction management, and modern mechanized tunnelling technologies. Throughout the construction process, careful planning, detailed surveys, continuous quality control, and strict safety procedures were implemented to ensure that the underground structures satisfied all design requirements while maintaining the safety of adjacent buildings and infrastructure.



### Project Location

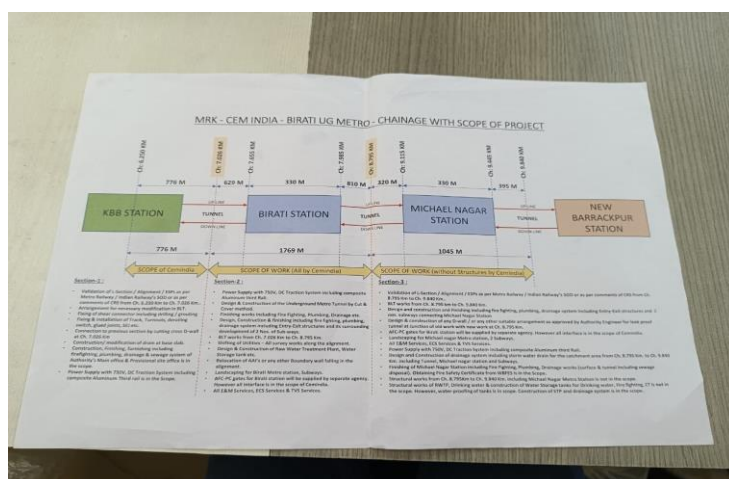
The Birati Underground Metro Station is situated in the North 24 Parganas district of West Bengal and forms part of the Kolkata Metro Railway expansion towards Barasat. The station has been strategically located to serve one of the rapidly developing suburban regions of Kolkata while providing efficient connectivity to the airport corridor and surrounding residential areas. The project alignment passes through densely populated urban zones where existing roads, commercial establishments, residential buildings, underground utility networks, and public infrastructure present significant engineering challenges during construction.

The underground section of the metro corridor extends between Birati and Michael Nagar. According to the construction drawings and site observations, important chainages were established throughout the project to identify station locations, tunnel sections, shafts, and

structural elements accurately. Chainage marking serves as the primary reference system during surveying, setting out, excavation, structural construction, and quality control activities. Every structural component, including diaphragm walls, sheet piles, raft foundations, tunnel boxes, roof slabs, and utility crossings, is positioned with respect to these predefined chainages to ensure construction accuracy.

The project site required extensive preliminary investigations before commencement of construction. Existing buildings, roads, drainage systems, electrical cables, water pipelines, sewer networks, communication lines, and other underground utilities were carefully mapped through detailed engineering surveys. Since the project is located within an urban environment, traffic diversion plans and public safety measures were also incorporated into the planning stage to minimize inconvenience during construction.

The proximity of the project to the airport region and other important public infrastructure demanded strict monitoring of vibration, settlement, and groundwater conditions throughout excavation and structural construction. These considerations significantly influenced the selection of excavation methods, retaining systems, and construction sequencing.



## Need for Underground Metro Construction

The rapid increase in urban population and private vehicle ownership has placed tremendous pressure on the existing transportation infrastructure of Kolkata. Roads in many parts of the city are already operating close to or beyond their design capacity, resulting in severe traffic congestion, increased travel time, fuel wastage, and environmental pollution. Expansion of road infrastructure in densely developed urban areas is often impractical due to limited land availability and the presence of existing buildings and public utilities. Consequently, underground metro systems provide an effective solution for enhancing transportation capacity without requiring extensive surface land acquisition.

Underground metro construction offers several engineering and operational advantages. By placing railway infrastructure beneath the ground surface, valuable urban land can be preserved for residential, commercial, and recreational purposes. Underground stations eliminate conflicts with road traffic, enabling uninterrupted train operations even during

peak traffic hours. Furthermore, underground metro systems improve passenger safety by separating railway operations from surface transportation and pedestrians.

Another significant advantage of underground construction is the reduction of environmental impacts. Metro systems consume less energy per passenger compared with private vehicles, thereby reducing greenhouse gas emissions and improving urban air quality. Underground alignments also minimize noise pollution because train operations occur below ground level. In addition, underground stations can be integrated with existing transport networks, including bus terminals, railway stations, airports, and commercial complexes, providing seamless multimodal transportation facilities.

From a structural engineering perspective, underground metro construction represents one of the most challenging forms of infrastructure development. Engineers must account for lateral earth pressure, groundwater pressure, surcharge loads from traffic and buildings, seismic forces, and long-term durability of reinforced concrete structures. Advanced construction techniques such as diaphragm walls, sheet piling, cut-and-cover excavation, Tunnel Boring Machines (TBMs), waterproofing membranes, and continuous instrumentation are therefore employed to ensure structural stability and public safety throughout the construction process.

### **Objectives of the Birati Underground Metro Project**

The Birati Underground Metro Project has been undertaken as a part of the expansion of the Kolkata Metro Railway network to address the rapidly increasing transportation demands of the northern metropolitan region. The project aims to provide a modern, reliable, efficient, and environmentally sustainable public transportation system while minimizing the adverse impacts of urban traffic congestion. The construction of the underground metro station and tunnel has been carefully planned to improve regional connectivity, enhance passenger convenience, and support the future development of the surrounding urban areas.

One of the primary objectives of the project is to reduce traffic congestion on the major roads connecting Birati, Michael Nagar, Jai Hind (Airport), and Barasat. The increasing number of private vehicles has placed considerable pressure on the existing road network, resulting in longer travel times, increased fuel consumption, and higher levels of air pollution. By providing a high-capacity underground metro system, the project significantly reduces dependency on road-based transportation and improves overall traffic management within the city.

Another important objective is to establish a safe and uninterrupted transportation corridor that remains operational regardless of weather conditions or surface traffic. Underground metro systems eliminate conflicts with road intersections, pedestrian crossings, and traffic signals, thereby ensuring reliable train operations with minimal delays. This contributes to improved passenger safety, operational efficiency, and service reliability.

The project also aims to enhance connectivity between important residential, commercial, educational, and transportation hubs. The Birati Underground Metro Station serves as a strategic link between Kolkata and the rapidly developing suburban areas of North 24

Parganas. Improved connectivity supports economic growth, encourages commercial development, and increases accessibility to employment opportunities, educational institutions, healthcare facilities, and other public services.

From an engineering perspective, the project seeks to implement advanced underground construction technologies capable of safely overcoming the challenges associated with deep excavation in densely populated urban areas. Modern construction methods such as the Cut-and-Cover Method, Tunnel Boring Machine (TBM) Method, diaphragm wall construction, sheet pile retaining systems, and reinforced concrete tunnel box construction have been adopted to ensure structural stability while minimizing settlement, vibration, and disturbance to nearby structures.

Environmental sustainability is another major objective of the project. Metro rail transportation consumes significantly less energy per passenger compared to private vehicles and produces substantially lower greenhouse gas emissions. The underground alignment further reduces noise pollution, preserves valuable urban land, and minimizes the need for large-scale demolition of existing buildings.

The project additionally aims to ensure long-term structural durability through the use of high-quality construction materials, advanced waterproofing systems, modern quality control procedures, and continuous monitoring during construction. Comprehensive instrumentation programs have been implemented to monitor settlement, groundwater levels, vibration, and structural movement, thereby ensuring the safety of both the construction site and surrounding infrastructure.

Overall, the Birati Underground Metro Project represents a significant milestone in the modernization of Kolkata's urban transportation system. Its objectives extend beyond transportation improvement by contributing to sustainable urban development, environmental conservation, public safety, and long-term economic growth.

#### **Scope of Work Executed by Cemindia Construction Pvt. Ltd.**

Cemindia Construction Pvt. Ltd. was entrusted with the execution of major civil and structural components of the Birati Underground Metro Project. The scope of work covered the complete construction of underground structural systems, including excavation support, reinforced concrete structures, waterproofing, entry and exit structures, utility coordination, and associated civil works required for the successful completion of the underground metro station.

The project commenced with extensive preliminary investigations, including detailed topographical surveys, utility surveys, geotechnical investigations, traffic studies, and structural condition assessments of adjacent buildings. These investigations were essential for understanding existing site conditions and identifying potential construction risks before excavation activities began.

Following the completion of survey and design activities, temporary earth-retaining systems were installed to facilitate deep excavation. Depending upon site conditions, diaphragm walls or Z-type steel sheet piles were used to retain the surrounding soil and prevent

collapse of the excavation. Steel walers and struts were subsequently installed at different levels to resist lateral earth pressure during excavation. These temporary support systems ensured stability until the permanent reinforced concrete tunnel structure became capable of carrying the earth pressure independently.

After excavation reached the required depth, the construction of the permanent underground structure commenced. This included laying of plain cement concrete (PCC), installation of waterproofing membranes, construction of raft foundations, reinforced concrete retaining walls, intermediate columns where required, roof slabs, and associated structural elements. The waterproofing system was carefully designed to prevent groundwater ingress and to ensure long-term durability of the underground structure.

The scope of work also included the construction of passenger entry and exit structures, staircases, lift shafts, escalator foundations, ventilation openings, drainage systems, service ducts, cable trenches, and other permanent civil works required for the operation of the metro station. Structural provisions were incorporated for the installation of Mechanical, Electrical, Plumbing (MEP), fire protection, signaling, communication, and ventilation systems.

In addition to structural construction, Cemindia implemented comprehensive quality assurance and quality control procedures throughout the project. Continuous testing of construction materials, inspection of reinforcement, monitoring of concrete quality, verification of structural dimensions, surveying of completed works, and monitoring of ground movement ensured compliance with project specifications and national standards.

Construction safety was given the highest priority throughout the project. Proper barricading, traffic management, confined space safety procedures, personal protective equipment (PPE), emergency response plans, and regular safety inspections were implemented to protect workers as well as the general public.

### **Construction Methodology Adopted**

The construction methodology adopted for the Birati Underground Metro Project was selected after detailed engineering investigations considering geological conditions, groundwater levels, traffic density, surrounding buildings, available working space, environmental impacts, and overall project economy. Since different portions of the alignment passed through varying site conditions, multiple construction techniques were employed to achieve maximum safety and construction efficiency.

The first stage of construction involved detailed engineering surveys and geotechnical investigations. Survey engineers established permanent reference benchmarks, control points, and tunnel centerlines using Total Stations, DGPS, GPS equipment, and precision leveling instruments. Existing roads, buildings, drainage systems, underground utilities, and property boundaries were accurately mapped before any construction activity commenced.

Following the completion of surveys, borehole investigations were carried out at regular intervals along the proposed alignment to determine soil stratification, groundwater conditions, bearing capacity, and engineering characteristics of the subsoil. Soil samples

collected from different depths were tested in laboratories to determine grain size distribution, moisture content, Atterberg limits, permeability, shear strength, consolidation characteristics, and other engineering properties. These investigations formed the basis for selecting suitable excavation methods, retaining structures, foundation systems, and waterproofing requirements.

After completion of design activities, temporary retaining systems were installed. In open areas where sufficient working space was available, the **Cut-and-Cover Method** was adopted. Steel sheet piles were driven into the ground using crawler cranes fitted with vibratory hammers. Excavation was carried out in stages, with steel walers and struts installed at different levels to support the sheet piles against increasing earth pressure. Once excavation reached the desired depth, a PCC bed was laid, waterproofing membranes were installed, and the permanent reinforced concrete raft foundation, walls, and roof slab were constructed. As the permanent structure gained strength, temporary struts were progressively removed until the complete tunnel box became self-supporting.

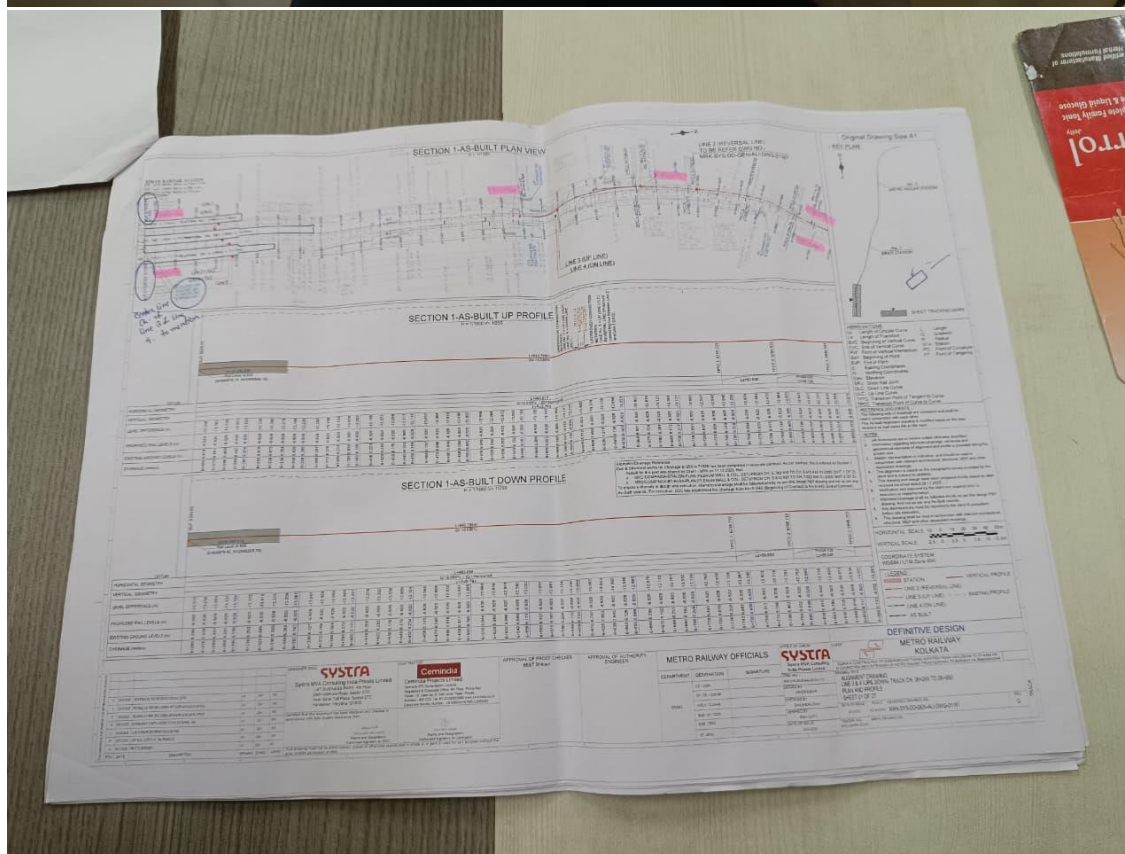
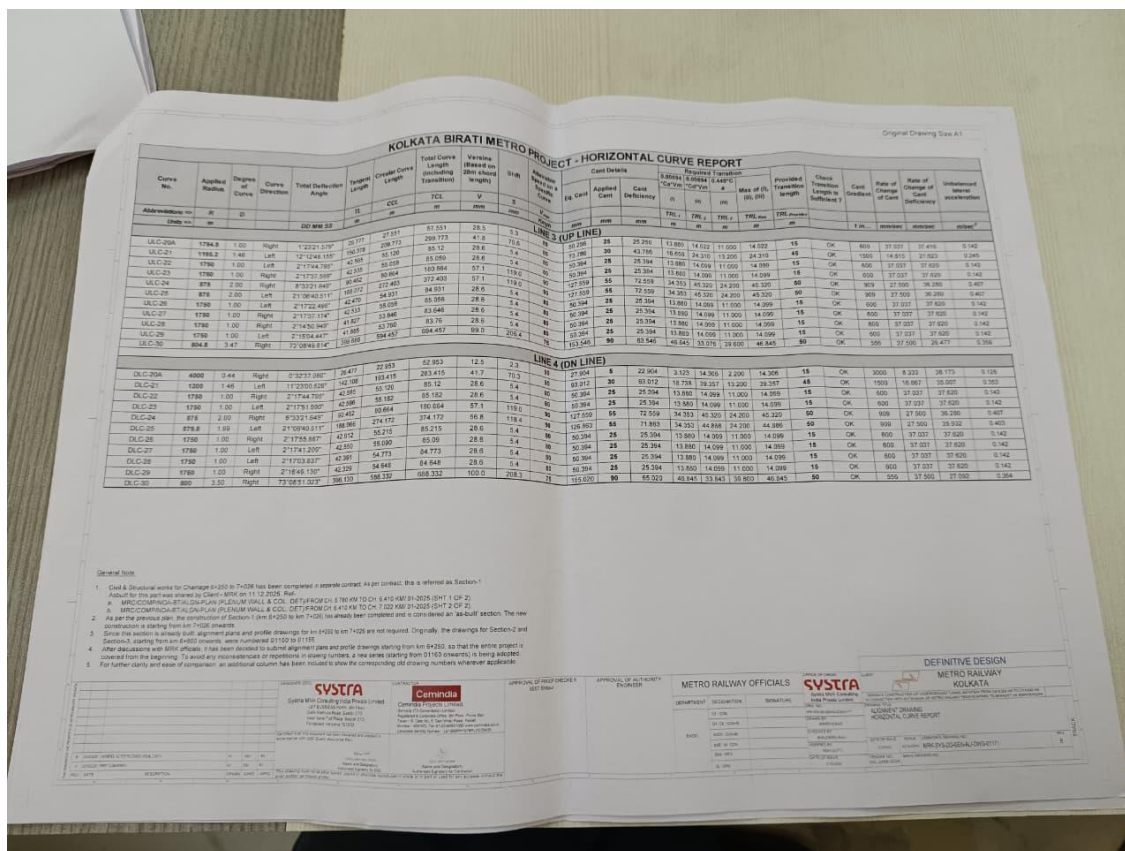
In densely populated areas with limited surface space and vibration-sensitive structures, the **Tunnel Boring Machine (TBM)** method was employed. The Earth Pressure Balance (EPB) TBM continuously excavated soil using a rotating cutter head while maintaining balanced pressure at the tunnel face to prevent ground settlement. Excavated soil was removed through a screw conveyor, and precast concrete tunnel lining segments were installed immediately behind the cutter head. The annular gap between the tunnel lining and surrounding soil was subsequently filled with grout to ensure stability and minimize settlement.

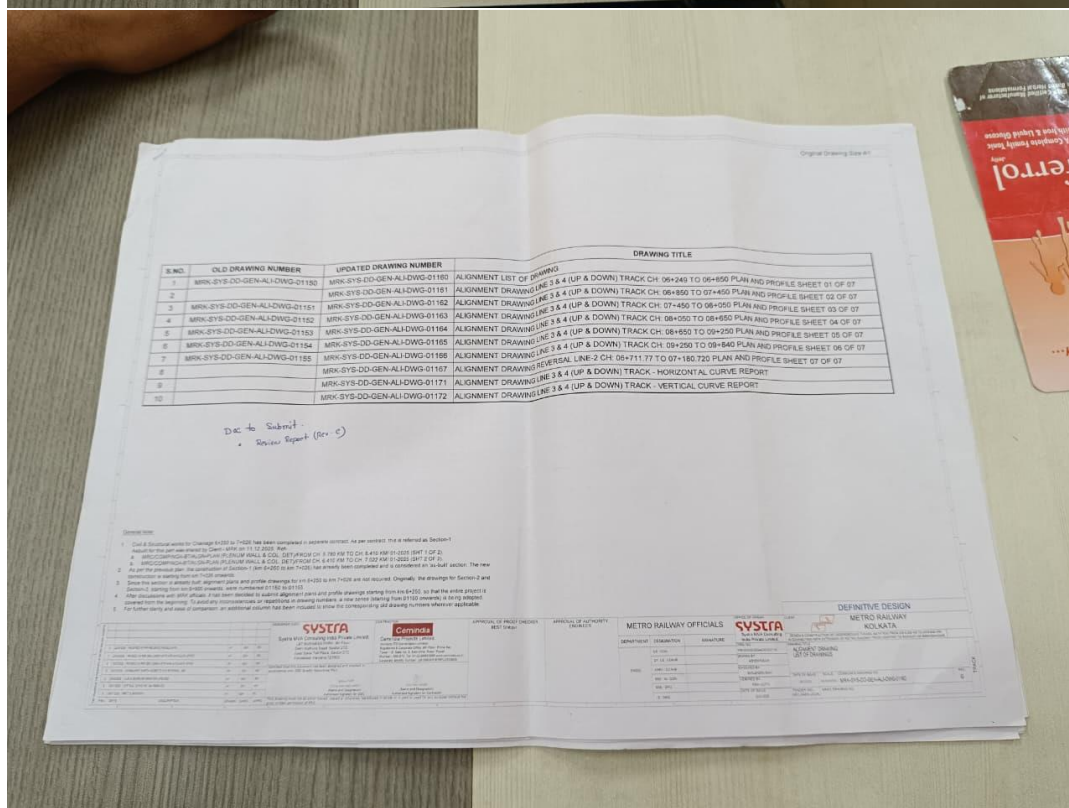
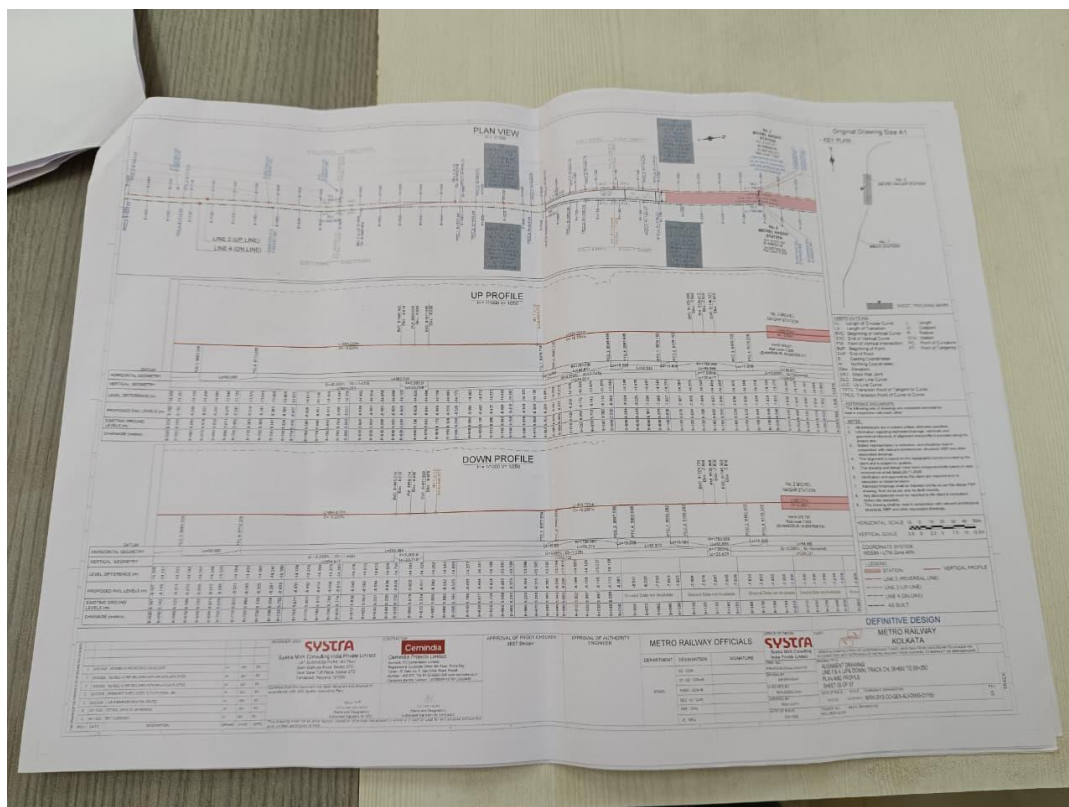
Throughout construction, continuous monitoring of settlement, vibration, groundwater levels, structural movement, and excavation stability was carried out using modern instrumentation. Quality control inspections and laboratory testing ensured compliance with project specifications, while strict safety measures protected workers and nearby residents during all phases of construction.

## **SURVEY AND GEOTECHNICAL INVESTIGATION**

This chapter presents the preliminary investigations carried out before the commencement of construction of the Birati Underground Metro Station and Tunnel. Surveying formed the foundation of the project by establishing the tunnel alignment, station locations, chainages, benchmarks, and control points required for accurate construction. Various surveying techniques, including reconnaissance survey, topographical survey, Total Station survey, DGPS survey, Auto Level survey, and utility mapping, were conducted to identify existing site conditions, surrounding infrastructure, and underground utilities. Geotechnical investigations were simultaneously performed through borehole drilling, soil sampling, laboratory testing, and groundwater analysis to determine the engineering characteristics of the soil. The information obtained from these investigations was used to finalize the alignment, select suitable construction methods, design retaining structures and foundations, and ensure safe execution of underground works. The chapter also discusses

traffic surveys, building condition surveys, vibration monitoring, and instrumentation adopted to minimize construction risks and protect nearby structures.





## **PROJECT ALIGNMENT AND DESIGN STAGES**

This chapter explains the planning and design process adopted for the Birati Underground Metro Project. It describes the alignment of the Yellow Line between Birati, Michael Nagar, Jai Hind (Airport), and Barasat, together with the chainage system used during construction. Based on the notebook sketches, the chapter also discusses the design workflow, including Preliminary Design (PD), Detailed Design (DD), Good for Construction (GFC) drawings, and As-Built (AB) drawings. Each stage of design played an important role in converting survey information into practical construction drawings and ensuring accurate site execution. The chapter further introduces the structural divisions responsible for underground structures, elevated structures, and diaphragm wall construction.

## **DIAPHRAGM WALL AND SHEET PILE CONSTRUCTION**

This chapter describes the temporary earth-retaining systems adopted during deep excavation. Depending on the site conditions, diaphragm walls and Z-type steel sheet piles were used to retain the surrounding soil and groundwater during excavation. The chapter explains the construction of guide walls, bentonite-supported trench excavation, reinforcement cage installation, tremie concreting, sheet pile driving using crawler cranes and vibratory hammers, and the installation of steel walers and struts to resist lateral earth pressure. Continuous monitoring of vibration, settlement, and wall movement ensured safe excavation without damaging nearby buildings or utilities. The engineering principles governing these temporary support systems are also discussed.

## **CUT-AND-COVER TUNNEL CONSTRUCTION**

This chapter presents the construction methodology adopted for the cut-and-cover tunnel sections between Birati and Michael Nagar. The complete sequence of construction, beginning with site preparation, surveying, excavation, sheet pile installation, temporary bracing, PCC laying, waterproofing, raft foundation construction, reinforced concrete wall construction, roof slab casting, backfilling, and road restoration, is explained in detail. The structural behaviour of the temporary and permanent components is discussed together with the engineering advantages and limitations of the cut-and-cover method. This chapter demonstrates how the permanent reinforced concrete tunnel gradually replaces the temporary support system during construction.

## **TBM TUNNEL CONSTRUCTION**

This chapter explains the mechanized tunnelling process adopted in densely populated urban areas where conventional excavation is not feasible. The working principle of the Tunnel Boring Machine (TBM), including the Earth Pressure Balance (EPB) system, rotating cutter head, screw conveyor, hydraulic jacks, segment erector, tunnel lining installation, and backfill grouting, is described. The chapter also discusses navigation systems, alignment control, tunnel lining assembly, and quality control measures adopted during TBM

operation. The notebook sketches showing the TBM components, EPB shield, and cutter head are referenced to explain the overall construction process.



**Fig : Tunnel Boring Machine (TBM) [Kolkata Metro]**

## **TEMPORARY AND PERMANENT WORKS**

This chapter distinguishes between temporary construction works and permanent structural components. Temporary works include sheet piles, diaphragm walls, steel walers, steel struts, dewatering systems, working platforms, and temporary drainage arrangements required during excavation. Permanent works include raft foundations, reinforced concrete tunnel walls, roof slabs, intermediate columns, tunnel boxes, entry and exit structures, and service ducts that form part of the completed metro station. The chapter explains how temporary systems ensure construction safety while permanent structures carry the design loads throughout the service life of the metro system.

## **WATERPROOFING, RAFT FOUNDATION, RCC WALLS, ROOF SLAB AND COLUMNS**

This chapter discusses the construction of the permanent reinforced concrete structural system after completion of excavation. Waterproofing membranes were installed above the PCC layer to prevent groundwater ingress and enhance the durability of the underground structure. Subsequently, raft foundations were cast to distribute structural loads safely to the supporting soil. Reinforced concrete walls, intermediate columns, and roof slabs were constructed according to the approved structural drawings. The structural functions of each component and the sequence of construction are explained together with reinforcement detailing, concreting procedures, curing, and quality control requirements.

## **ENTRY & EXIT STRUCTURES, MEP SERVICES, ARCHITECTURAL WORKS AND TRACK WORKS**

This chapter describes the finishing and functional components of the underground metro station after completion of the primary civil structure. It includes the construction of passenger entry and exit structures, lift shafts, escalator foundations, ventilation systems, fire protection installations, electrical services, plumbing, drainage, communication systems, architectural finishes, and railway track installation. The integration of these systems transforms the completed structural shell into a fully operational underground metro station capable of safely serving passengers.

## **MONITORING, INSTRUMENTATION, SAFETY AND QUALITY CONTROL**

This chapter explains the monitoring systems adopted throughout construction to ensure structural safety and quality. Settlement markers, inclinometers, piezometers, vibration meters, crack meters, and other instrumentation were installed to monitor ground movement, groundwater levels, structural displacement, and vibration during excavation and sheet pile installation. The chapter also describes construction safety measures, including personal protective equipment (PPE), confined space procedures, emergency response systems, traffic management, and routine safety inspections. Quality control activities such as material testing, reinforcement inspection, concrete testing, waterproofing inspection, and dimensional verification are discussed to demonstrate compliance with engineering standards.

## **CONSTRUCTION SEQUENCE, AS-BUILT DRAWINGS, STRUCTURAL OBSERVATIONS AND CONCLUSION**

The final chapter summarizes the complete construction sequence of the Birati Underground Metro Project, beginning with survey and geotechnical investigations and concluding with commissioning of the completed underground station. It explains the preparation and importance of As-Built drawings, which record the actual dimensions, levels, and modifications executed during construction. Based on the site visit and notebook observations, the chapter presents key structural observations related to tunnel construction, diaphragm walls, sheet piles, TBM operations, waterproofing, raft foundations, and reinforced concrete works. Finally, the report concludes by highlighting the engineering knowledge gained during the internship, the practical application of surveying, geotechnical engineering, structural engineering, and underground construction techniques, and the successful execution of the Birati Underground Metro Project by Cemindia Construction Pvt. Ltd. This chapter emphasizes how the project integrates modern construction technology, quality assurance, safety management, and multidisciplinary coordination to deliver a durable and efficient urban transportation system.

# Diaphragm Wall

## ( D - W A L L )

### 1. What is a Diaphragm Wall?

A Diaphragm Wall (D-Wall) is a reinforced concrete structural wall cast in-situ within a narrow trench excavated in the ground. The trench is stabilised throughout excavation by filling it with a polymer slurry whose hydrostatic pressure counteracts lateral earth and groundwater pressure, preventing collapse.

Once the required depth is reached, a pre-fabricated reinforcement cage is lowered into the slurry-filled trench and concrete is placed from the bottom up using tremie pipes, displacing the slurry and forming a monolithic wall panel. Sequential panels, connected through watertight joints, form a continuous perimeter wall.

The D-Wall serves a dual function — it acts as the temporary retaining structure during excavation and as the permanent structural wall of the underground station box, making it the most economical solution for deep urban construction.

### Key Specifications

| Parameter         | Specification                                     |
|-------------------|---------------------------------------------------|
| Wall Thickness    | 1,000 mm (1.0 m)                                  |
| Concrete Grade    | M 40/20   Slump: 150–175 mm                       |
| Reinforcement     | Fe 500D / 550D (HYSD TMT, IS:1786)                |
| Stabilising Fluid | BT Polymer Slurry (PHPA-based)                    |
| PVC Water Stop    | 10 mm thick x 230 mm wide, 2-layer at every joint |

### 2. Working Principle

The fundamental principle relies on the hydrostatic pressure of polymer slurry maintaining trench stability against surrounding soil and groundwater during excavation. The slurry

column is kept above the water table at all times to ensure a net positive inward pressure on the trench walls.

During concreting, the heavier concrete ( $\approx 24 \text{ kN/m}^3$ ) displaces the lighter slurry ( $< 1.025 \text{ g/cc}$ ) progressively from the bottom upward through the tremie method, forming a dense, defect-free monolithic wall panel.

### 3. Step-by-Step Construction Procedure

#### **Step 1 — Survey & Setting Out**

Precise survey establishes the exact D-Wall alignment. Reference datums and alignment lines are marked on-site, and panel numbers are identified on the guide wall top in accordance with approved drawings before any excavation begins.

#### **Step 2 — Guide Wall Construction**

Inverted L-shaped, in-situ concrete guide walls are constructed at ground level on both sides of the D-Wall alignment. They control alignment of the grab, support trench walls at surface, and guide the reinforcement cage into position.

| Parameter              | Specification                            |
|------------------------|------------------------------------------|
| Shape                  | Inverted L-shaped, cast in-situ concrete |
| Min. clear gap         | D-Wall thickness (1,000 mm) + 25 mm      |
| Max. clear gap         | D-Wall thickness (1,000 mm) + 50 mm      |
| Verticality tolerance  | $\leq 15 \text{ mm per } 3 \text{ m}$    |
| Construction tolerance | $\pm 50 \text{ mm}$                      |

#### **Step 3 — Panel Sequencing**

The wall is built panel by panel in a defined P–S–S–C sequence to maintain trench stability and achieve watertight joints. Panel length is typically 5.4 to 6.0 m.

| Panel Type           | Role                                                           | Stop-Ends      |
|----------------------|----------------------------------------------------------------|----------------|
| <b>Primary (P)</b>   | First panel cast — starting point, proceeds in both directions | 2 (both sides) |
| <b>Secondary (S)</b> | Cast adjacent to primary panels                                | 1 (one side)   |
| <b>Closing (C)</b>   | Final panel — after removal of both stop-ends                  | None           |
| <b>End (E)</b>       | Wall extremity — special configuration                         | As per drawing |

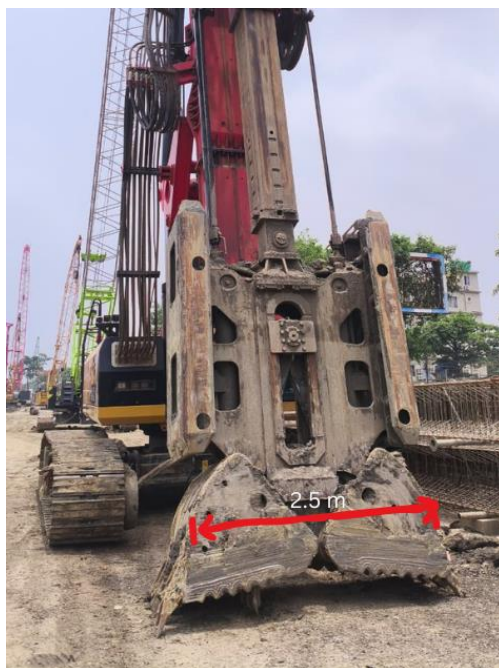
#### **Step 4 — Trench Excavation (Hydraulic Clamshell Grab)**

A hydraulic clamshell grab mounted on the rig (SR-235 / HR-180) bites soil in repeated passes under polymer slurry until design depth is reached. The grab is continuously monitored for verticality and any deviation corrected immediately.

- \* Koden Drilling Monitor (DM-604) provides real-time verticality measurement
- \* Rig onboard digital display shows live inclination angle, excavation depth and grab load
- \* Slurry level maintained above water table throughout



**Fig: Hydraulic clamshell grab rig (SR-235 type)**



**Fig: Grab bucket close-up — 2.5 m cutting width (bites soil in repeated passes)**

### **Step 5 — Polymer Slurry (Stabilising Fluid)**

BT Drilling Polymer (PHPA-based, dry granular form) is used instead of conventional bentonite slurry. Mixed at  $1 \text{ kg/m}^3$  with continuous air agitation, it produces a clear, high-viscosity stabilising fluid that leaves cleaner concrete surfaces, higher rebar bond strength, and improved recyclability over bentonite.

| Parameter        | Specification                               | Test Method      |
|------------------|---------------------------------------------|------------------|
| <b>Viscosity</b> | 34–143 sec/Litre (site: 50–100)             | Marsh Cone (API) |
| <b>pH</b>        | 8.0–11.0 (corrected with NaOH / KOH)        | pH Paper / Meter |
| <b>Density</b>   | < 1.025 g/cc                                | Hydrometer       |
| <b>Mix Ratio</b> | $1 \text{ kg/m}^3$   Mixing time: 30–60 min | —                |

| Parameter       | Specification                         | Test Method |
|-----------------|---------------------------------------|-------------|
| Recycled Slurry | 70–80% reused; 0.5% BT polymer top-up | —           |

### **Step 6 — Stop-End & Water Stopper**

Peel-off type stop-ends are installed at panel ends before concreting to form the joint face and carry the PVC water stopper (10 mm x 230 mm wide, 2-layer). After the adjacent panel is excavated, the stop-end is extracted by a chisel-and-clamp 'peel-off' technique.

### **Step 7 — Reinforcement Cage**

Cages are fabricated using Fe500D/550D TMT bars, GI binding wire (18 SWG), and welding. They incorporate all main bars, shear links, mechanical couplers at slab junction levels, and spacer bars as per the approved structural drawing.

| Element                    | Specification                                    |
|----------------------------|--------------------------------------------------|
| Main bars — Soil face      | T32, T25, T20 — bundled bar layers               |
| Main bars — Excavated face | T25, T20, T16                                    |
| Shear links                | T8, T10 @ 100–150 mm c/c                         |
| Coupler bars               | T25, T20 @ 135 mm c/c — 2nd layer at slab levels |
| Spacer (Z-bar)             | T32 @ 280 mm c/c                                 |
| Cover — bottom             | 75 mm                                            |
| Cover — top surface        | 50–55 mm                                         |
| Cover block spacing        | 1,500 mm horiz. x 4,000 mm vert.                 |



**Fig: Rebar with couplers — ready for cage fabrication**



**Fig: Shear-link / stirrup cages laid out in sequence for main-bar assembly**



**Fig: Cover blocks fixed on soil-face bars — 1,500 mm horiz. x 4,000 mm vert. spacing**



**Fig: Close-up of tied rebar joints and welded lacer-bar connections**



**Fig: Assembled reinforcement mat on the fabrication bed, cover blocks fixed, prior to rotation and lifting**



**Fig: Threaded mechanical coupler sleeve (close-up)**

## **Mechanical Couplers**

Pre-embedded couplers (16, 20, 25 and 32 mm dia.) are installed at slab junction levels — Top of Box, Concourse, Undercroft and Toe. They allow internal slabs to be connected to the D-Wall after excavation. Thermocol lagging protects them inside the wall until chipping.

Coupler Tolerance (approved drawing): Longitudinal  $\pm 75$  mm | Vertical  $\pm 50$  mm. Reducer couplers used at bar diameter transitions (e.g., T32  $\rightarrow$  T25).

## **Cage Lifting & Lowering**

- \* 22.5 m cage, ~22.5 MT — tandem lift by two crawler cranes
- \* Crane-A (150 MT) at 10.0 m radius — lowers cage solo into trench
- \* Crane-B (80 MT) at 8.0 m radius — supports during initial horizontal-to-vertical rotation
- \* Tolerances: Longitudinal  $\pm 75$  mm | Vertical  $\pm 1:200$  | Lateral  $\pm 25$  mm

Crane safety check (Annexure B — Cage Lifting & Lowering Plan): Crane-A (150 MT) at 10.0 m radius, 40 m boom — capacity 50.0 MT @75%, load transferred 5.0 MT — Safe. Crane-B (80 MT) at 8.0 m radius, 30 m boom — capacity 25.0 MT @75%, load transferred 17.5 MT — Safe.



**Fig: Cage being tandem-lifted — 22.5 m length, 3.9 m panel width at top and 0.8 m at lifting yoke**



**Fig: Reinforcement cage lowered and positioned inside the slurry-filled trench, guided by the guide walls**

### **Step 8 — Tremie Concreting**

Concrete (M40/20, slump 150–175 mm) is placed through 200/250 mm diameter steel tremie pipes from the base of the trench upward, continuously displacing slurry.



**Fig: Tremie pipe discharging concrete through the hopper into the trench**

1. Vermiculite plug inserted into tremie hopper before first discharge — prevents concrete-slurry mixing
2. Multiple tremie pipes ensure uniform concrete rise across the panel width
3. Minimum 1.5–2.0 m tremie immersion in rising concrete maintained at all times
4. Concrete poured above cut-off level — excess chipped away after setting
5. Adjacent panel excavation not permitted within 12 hours of pour completion

#### **4. Verticality Monitoring — Kodon Drilling Monitor (DM-604)**

The Kodon DM-604 is an ultrasonic echo-meter that measures trench wall profiles in real-time. A sensor probe lowered into the slurry-filled trench emits ultrasonic pulses that reflect off the trench walls; the time-of-flight of each echo is converted to distance, producing a continuous cross-sectional plot of trench width and verticality in both X and Y axes as the probe descends.

- \* Any deviation from vertical is immediately corrected by rotating and chopping the grab
- \* The rig's onboard digital control panel cross-checks live inclination angle and depth against the Kodon printout



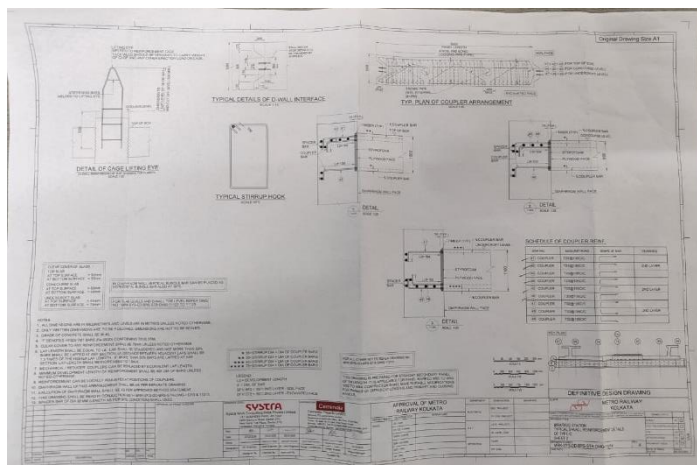
**Fig: Kodon DM-604 with printed verticality chart**



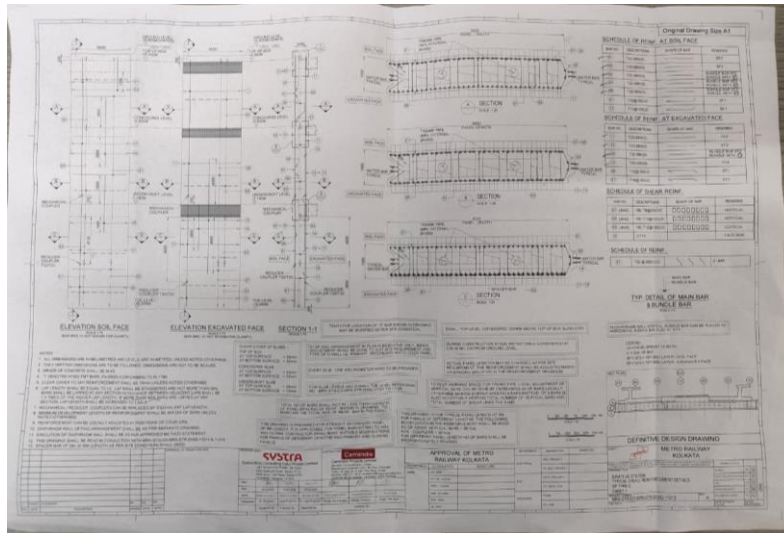
**Fig: 6 Rig display**

## 5. Approved Structural Reinforcement Drawings

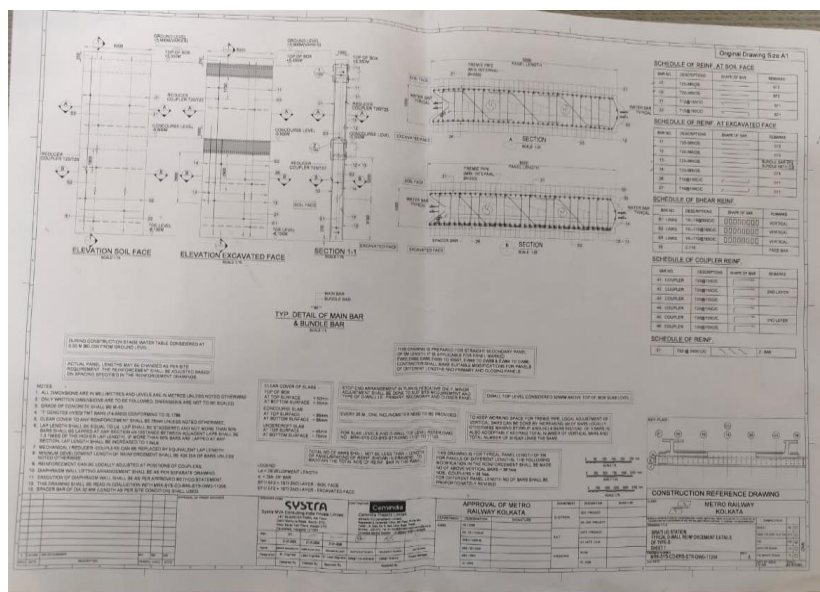
Definitive Design and Construction Reference drawings for Birati Station D-Wall reinforcement, prepared by SYSTRA and executed by Cemindia Projects Limited for Metro Railway, Kolkata.



**Fig: Elevation (Soil Face & Excavated Face), Section 1-1, reinforcement schedules and bundle bar detail**



**Fig: Coupler arrangement plan at Top of Box & Concourse Level, Cage Lifting Eye detail, D-Wall Interface details**



**Fig: Stirrup hook detail, coupler plan, D-Wall interface sections and Schedule of Coupler Reinforcement**

## 6. Top-Down vs Bottom-Up Construction

**Comparison Table: Top-Down vs. Bottom-Up Construction**

| Feature / Step                        | Top-Down Method                                                                                                           | Bottom-Up Method                                                                                                                             |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Initial Phase</b>               | Surveying                                                                                                                 | Surveying                                                                                                                                    |
| <b>2. Support Structure</b>           | Installation of a permanent retaining wall (e.g., D-Wall, Sheet Pile, or Soldier Pile).                                   | Installation of a retaining wall (D-Wall, Sheet Pile, or Soldier Pile) which initially requires temporary support.                           |
| <b>3. First Excavation</b>            | Initial excavation is done to casting level.                                                                              | Excavation is carried out down to the 1st strut level only.                                                                                  |
| <b>4. Upper Slab / Support</b>        | The Roof Slab is cast immediately after the initial excavation, acting as permanent bracing.                              | A temporary strut/support (1st layer) is installed instead of a permanent slab.                                                              |
| <b>5. Subsequent Excavation</b>       | Further excavation continues downward underneath the cast roof slab through temporary openings until the base is reached. | Excavation continues downward sequentially in stages (e.g., up to the 2nd layer, then 3rd layer), installing temporary struts at each level. |
| <b>6. Structural Elements</b>         | Utilizes elements like a Plunge Column to support downward loads during construction.                                     | Structure is built conventionally from the Base Slab upwards once full depth is reached.                                                     |
| <b>7. Structural Components Shown</b> | Roof Slab, Concourse Slab, Base Slab, and Plunge Column.                                                                  | Struts handling lateral and axial loads, running alongside Cut & Cover / TBM work.                                                           |

## 7. Defect Prevention

### Verticality Control

Koden DM-604 monitors trench verticality continuously during excavation. Deviations are corrected immediately by rotating the grab. Post-excavation bulges are chipped off using pneumatic hammers to match the design alignment.

### Joint Water-Tightness

Water seepage through joints, if observed, is remedied by chipping joints to 50 mm depth, grouting with cement-grout mix, and repairing with suitable mortar. The embedded PVC water stopper provides the primary waterproofing.

## 8. Major Plant & Equipment

| Equipment                                      | Qty. | Purpose                         |
|------------------------------------------------|------|---------------------------------|
| Hydraulic Rig SR-235 / HR-180 with D-Wall Grab | 2    | Trench excavation               |
| Crawler Crane 150 MT (Crane-A)                 | 1    | Primary cage lowering crane     |
| TFC 280 Crawler Crane 75 MT (Crane-B)          | 2    | Supporting crane / tandem lifts |
| Koden Drilling Monitor DM-604                  | 1    | Trench verticality measurement  |
| Positive Displacement Pump                     | 2    | Polymer slurry circulation      |
| 100 MT Hydraulic Jack + Power Pack             | 1    | Stop-end extraction             |
| Bar Cutting Machine (Spartan C-52)             | 4    | Reinforcement cutting           |
| Bar Bending Machine (B52)                      | 4    | Reinforcement bending           |
| Welding Rectifier 400 Amp                      | 12   | Cage assembly                   |
| Air Compressor 365 CFM                         | 1    | Slurry agitation                |
| Vertical Pump 15HP / 20HP                      | 6    | Slurry pumping                  |
| Electric Concrete Vibrators 3HP                | 2    | Concrete compaction             |

## 9. Site Observations & Key Learnings

| Observation                                     | Significance                                                                                                |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Polymer slurry over bentonite</b>            | Cleaner surfaces, higher rebar bond, no contamination of concrete — superior for permanent walls            |
| <b>Koden + rig screen together</b>              | Dual real-time verification of verticality — any deviation caught and corrected in the same pass            |
| <b>Couplers pre-embedded in cage</b>            | Placement precision ( $\pm 75$ mm / $\pm 50$ mm) determines quality of slab-to-D-Wall connection            |
| <b>P→S→C panel sequence</b>                     | Prevents fresh concrete adjacency — avoids joint defects and maintains wall continuity                      |
| <b>Tandem crane lift</b>                        | 22.5 m cage requires two cranes and a detailed Cage Lifting Plan as critical as the Method Statement itself |
| <b>Tremie immersion <math>\geq 1.5</math> m</b> | Single most important factor preventing slurry inclusion inside the wall                                    |
| <b>Top-Down adopted at Birati</b>               | Early surface restoration — essential in a dense urban corridor with live traffic                           |

## POLYMER

### What is Polymer?

Polymer is a water-soluble vinyl polymer in an innovative easy mixing, dry granular form. Polymer is designed for preparation of viscous earth-stabilizing fluids for a variety of excavation techniques including drilling, trenching and walling applications in the geo-construction industries. For Preparation of Polymer Slurry as drilling fluid, following components of Polymers to be mixed.

While constructing a diaphragm wall (D-wall) for an underground metro project, excavation stability is everything. In deep trenching operations for the Kolkata Metro, the trenches are incredibly deep and prone to collapsing from the lateral pressure of the surrounding soil and groundwater.

To prevent this, the trench is kept continuously filled with a heavy, viscous fluid called **drilling slurry** or **stabilizing fluid**. While bentonite clay has historically been the industry standard, **synthetic polymers** (like the water-soluble vinyl polymers mentioned in your document) have largely taken over modern urban metro projects.

### How Polymer Slurry Works in D-Wall Construction

Polymer slurries stabilize trenches through a completely different mechanism than bentonite clay:

- **The Hydrostatic Push:** Like bentonite, the weight of the fluid column exerts a positive outward pressure against the trench walls, counteracting the active earth and groundwater pressure.
- **Chemical Binding (The "Sticky" Web):** Instead of forming a thick mud cake on the trench wall, long-chain polymer molecules penetrate slightly into the surrounding soil. They physically wrap around and chemically bind the individual soil particles together, forming a highly stable, cohesive matrix along the excavation interface.
- **Viscosity Control:** The polymer chain creates high viscosity under low-shear conditions (when the fluid is just sitting in the trench, holding it open), but thins out dramatically under high-shear conditions (when a grab is actively digging).
- 

### Why Metro Projects Prefer Polymer over Bentonite?

For a dense urban metro environment (like the Birati/Michael Nagar stretch in Kolkata), polymers offer massive logistical and technical advantages over traditional bentonite muds:

| Feature                             | Bentonite Slurry                                                                                                   | Polymer Slurry                                                                                                                             |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Footprint on Site</b>            | Requires huge desanding plants, massive mixing tanks, and significant real estate.                                 | Compact mixing units. Requires a fraction of the storage space because much less material is needed.                                       |
| <b>Structural Integrity</b>         | Forms a thick "filter cake" on the trench wall that can sometimes entrap mud and weaken the concrete-to-soil bond. | Leaves no thick filter cake. It rinses off cleanly when concrete is poured, leading to <b>higher skin friction</b> and a cleaner D-wall.   |
| <b>Environmental &amp; Disposal</b> | Difficult and expensive to dispose of. It becomes a heavy, unmanageable cake of contaminated mud.                  | Can be chemically broken down (e.g., using household bleach/hypochlorite) back into water and clean sand, making disposal eco-friendly.    |
| <b>Recyclability</b>                | Degrades quickly when contaminated with fine silts, requiring constant replacement.                                | Excellent settling properties; sand and silt drop rapidly to the bottom of the settling tanks, allowing the fluid to be reused many times. |

## METHOD STATEMENT FOR POLYMER MIXING

### 1. References

I. Contract Documents

II. Quality Assurance manual.

III. Quality system manual.

IV. Health, safety and Environmental plan

### 2. Method statement for polymer mixing

(i). BT Drilling Polymer.

(ii). Sodium Hydroxide or Potassium Hydroxide.

(iii). BT ALPHA BOND.

(iv). BT Micro Bond.

## CHEMICAL AND PHYSICAL CHARACTERISTICS OF DRY BT POLYMER

|                       |                       |
|-----------------------|-----------------------|
| <b>Physical State</b> | White Granules Powder |
| <b>Charge</b>         | Anionic               |
| <b>Density</b>        | 1.00 – 1.025 gm/cc    |

### Description :-

BT drilling Polymer is A free flowing, Water-soluble, easy, mixing 100% active, dry granular polymer. BT drilling Polymer Stabilizing agent is a very high molecular weight partially mix of hydrolysed polyacrylamide (PHPA) polymer and low viscous PAC. When mixed with fresh water, a small quantity of BT drilling Polymer Stabilizing agent can provide a clear, Solid- free, viscous borehole stabilizing fluid for used in drilled shaft, auger drilling, horizontal directional boring, trenching excavation and reverse circulation (RC) rotary drilling. BT drilling Polymer is not designed to be used in conjunction with bentonite-based fluids.

### Applications/Functions

Use the BT drilling Polymer dry polymer assists or promotes the following:-Build a clay free boring fluid. Stabilizer reactive slay and shale formations. Enhance core recovery in continuous wireline coring operations. Provide high cohesiveness. to bind excavated sandy soil and gravel. Removal of drilled spoils from augers and increased excavation.

### 3. Standard operation Procedures for Drilling Fluid

#### Material required for Best performance of BT drilling Polymer:-

- Polymer Drilling Fluid
- Mixing tanks minimum capacity as per site requirement.
- Storage tank minimum capacity as per site requirement.
- Sedimentation tank if needed with capacity as per site requirement.
- Air compressor for good agitation of polymer slurry during mixing.
- Pumps for sending and retrieval of polymer slurry.
- Sodium hydroxide (NaOH) or Potassium Hydroxide (KOH) for pH correction.
- Sodium Silicate and Sodium Aluminates to prevent fluid loss and continuous collapse if required.
- Testing kit (Mars cone, pH paper/meter, Hydro meter).

### pH Correction :

The pH of the drilling slurry can affect performance of the drilling slurry. Slurry will have maximum hydration where the pH is between 8 and 11. Sodium Hydroxide (NaOH) or Potassium Hydroxide (KOH) with 50% concentration should be mix with the ratio of 1L/m<sup>3</sup> to raise the pH up to its required limit.

### Mixing and hydration of Polymer Slurry:-

Appropriate water supply should be assured all the times in working site, in case of insufficient water supply additional storage tanks dedicated to water storage should be available in the working site.

Excavation should be done with clean drilling fluid, slurry should be prepared with the ratio of 1 kg/m<sup>3</sup> for the fresh mix, under constant air agitation provided by air compressor and circulation of slurry should be done from one end to another in the mixing tank by means of positive displacement Pump, a metallic plate should be welded directly underneath the flow of water directly in to mixing tank with an Inclination of 45 degree in order to get faster hydration.

After the fluid is mixed, sufficient time of minimum 2 hours must be allowed to Elapse to insure complete hydration of the Polymer prior to it being circulated into the pile Bore or Diaphragm wall panel. In case of brackish or salt water pre treatment of water should be done as per instructed by BT Polymer's Technical Consultant.

Parameters should be mentioned all time before sending the slurry to pile bore hole or Diaphragm wall panel, after excavation and after placing the reinforcement cage.

### Parameter Table For Testing

| Parameter | Fresh Mix       | Re-cycled       | Prior to concrete pour | Apparatus to examine sample |
|-----------|-----------------|-----------------|------------------------|-----------------------------|
| Viscosity | 34–143 sec      | 34–143 sec      | 34–143 sec             | Marsh cone (API)            |
| pH        | 8–11            | 8–11            | 8–11                   | pH Paper/Meter              |
| Density   | 1.025 g/cc max. | 1.025 g/cc max. | 1.025 g/cc max.        | Hydrometer                  |

### Storage tanks and Sedimentation Tank :

Storage Tanks or silos are used to store slurry prepared for excavation as well as water storage. Sedimentation Tank volume should be calculated in order for sedimentation to occur effectively, sedimentation tank should be open from the top of the tank.

### Positive Head pressure:

A minimum height of 2 meters of slurry column above the water table should be maintained in order to get positive differential head pressure, plans for design and building of working platform should take this factor in to consideration.

### **Drilling Tools:**

Drilling Tool must have drainage holes of above **100 mm** to receive slurry when extracted from excavation to prevent fluid loss due to tool action; Drilling tools must have inflow vents to minimize suction effect.

### **Supply of Polymer to boring location:**

After preparation of solution the same is to be transferred to the boring and the supply to be continued until the boring completed. Similarly, once the concrete has started to discharge the solution shall be taken back progressively and stored in sedimentation tank for further treatment and recycling.

### **Sampling of Polymer slurry:**

Volume of each sample should be sufficient to perform viscosity, density, and pH value test. Samples should be taken during excavation after completion of excavation and after placing rebar cage in the pile bore or diaphragm wall panel.

## **4. SUMMARISED MIXING PROCEDURE:-**

### **(a) Fresh Mix:-**

|                                                                        |
|------------------------------------------------------------------------|
|                                                                        |
| Sodium Hydroxide-1ltr/cum (to increased the PH Value of fresh Water)   |
| Polymer for fresh mixing = 1kg/cum (for 60-100 viscosity)              |
| API specification: -viscosity (34 to 143)                              |
| At site viscosity = 50-100 (as per site condition).                    |
| Mixing time: -30minutes to 60 minutes                                  |
| BT ALFA Bond to be mixed with Polymer Slurry @0.50kg/cum for Fresh Mix |

### **(b) Returned Polymer:-**

|                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Micro bond (BT Micro Bond)-1% of polymer slurry to sedimentation as per site requirement.                                                                                               |
| Approx. 70%-80% of used Polymer Slurry will be returned to the sedimentation Tank and 0.5% of BT polymer will be added for re- mixing for maintain the viscosity as per site condition. |

## **MARSH CONE TEST OF POLYMER SLURRY:-**

The Marsh Cone Test is a field quality control test used to determine the apparent viscosity (flowability) of the polymer slurry used to stabilize the trench during diaphragm wall construction. Proper viscosity ensures that the slurry can support the trench walls, suspend excavated soil particles, and allow efficient concrete placement through the tremie method. The test is widely used as a routine quality-control method for construction slurries

### **Objective**

- To measure the viscosity of polymer slurry.
- To ensure the slurry has adequate trench-supporting capacity.
- To verify that the slurry satisfies project specifications before excavation and before

### **Apparatus**

- Standard Marsh cone (Marsh funnel)
- 946 mL (1 quart) calibrated measuring cup
- Stopwatch
- Stirring rod
- Clean collection container

### **Test Procedure**

1. Mix the polymer slurry thoroughly.
2. Close the outlet of the Marsh cone with a finger or stopper.
3. Fill the cone with slurry up to the specified level.
4. Place the calibrated 946 mL cup below the outlet.
5. Release the outlet and start the stopwatch simultaneously.
6. Stop the stopwatch when 946 mL of slurry has been collected.
7. Record the flow time in seconds, known as the Marsh Cone viscosity.



**Fig: Marsh Cone Test**

**Observation:**

| Parameter            | Observation                                                                     |
|----------------------|---------------------------------------------------------------------------------|
| Total Volume         | 1500 mL                                                                         |
| Volume collected     | 946 mL                                                                          |
| Flow time            | 2 minutes 54 seconds = 174 Seconds                                              |
| Marsh Cone viscosity | $1.025 \times (174-25) = 152.725 \text{ cp (centi poise)} = 0.152 \text{ Pa-s}$ |

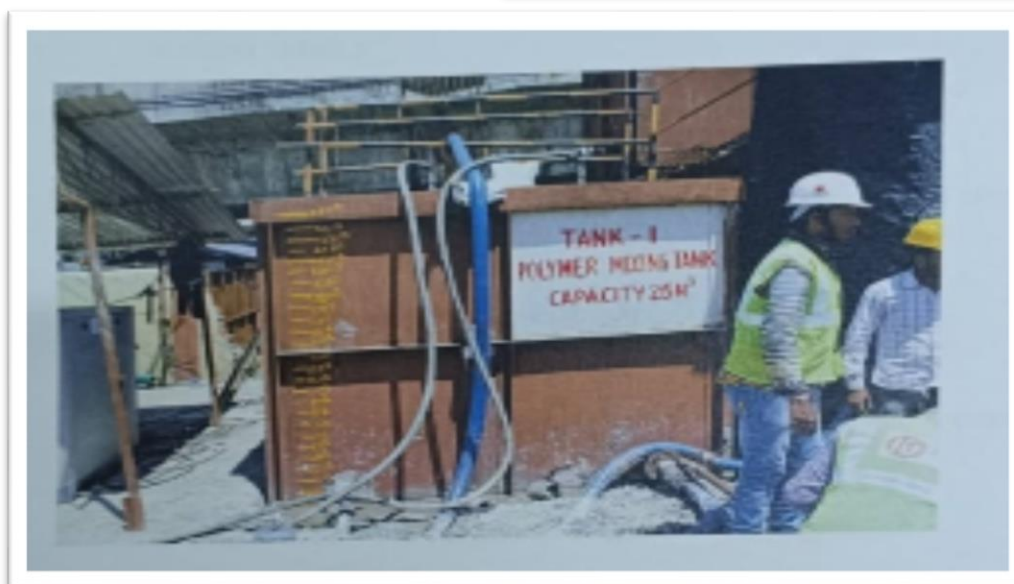
**Importance of this test in Diaphragm Wall Construction**

- Maintains trench stability and prevents sidewall collapse.
- Suspends excavated soil particles effectively.
- Minimizes soil settlement around nearby structures.
- Allows proper displacement by concrete during tremie concreting.
- Produces a sound, defect-free diaphragm wall.

## Precautions

- Mix the slurry thoroughly before testing.
- Remove large debris that could block the funnel outlet.
- Perform the test at a consistent temperature.
- Clean the Marsh cone after each test.
- Always compare results with the approved project specification and polymer manufacturer's recommendations.

5. Some site photographs depicting the schematic arrangement of polymer mixing and supplying:-





### **Merits of Polymer over bentonite:**

1. Less equipment needed for slurry operations (No desander required & heavy pump).
2. Reduce on site stock (proportion may be less than 50 times to betonites).
3. Greater frictional load bearing capacity (skin friction) in bored piles, higher success rate on load tests.
4. Significantly increased concrete to rebar bonding capacities.
5. Cleaner, harder top surfaces on concrete in bored piles and diaphragm walls; virtually eliminate cleaning, chipping and dressing.
6. Cleaner, smoother exposed diaphragm wall surfaces.
7. Tighter joints between wall panel improved water tightness.
8. Improved recyclability and reusability.
9. Cleaner and more manageable site.
10. Simplified disposal.
11. No contaminated concrete.

## Sheet Pile Installation

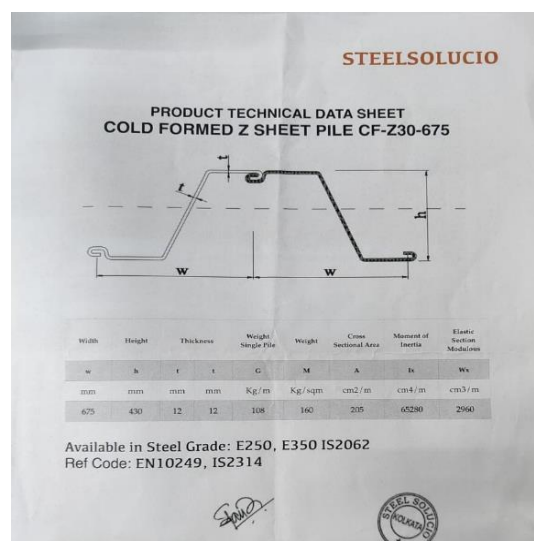
### Birati Underground Metro Project (Birati – Michael Nagar Corridor)

#### Project Overview

The Birati Underground Metro Project is a major urban transportation infrastructure project extending from Birati to Michael Nagar. The project involves the construction of underground metro stations and tunnels in densely populated urban areas where maintaining excavation stability and minimizing disturbance to surrounding structures are of paramount importance.

The project is being executed by Cemindia. Due to the presence of soft alluvial soil, a high groundwater table, and adjacent buildings and roadways, steel sheet pile walls were adopted as the primary temporary earth-retaining system before deep excavation.

In construction of the Birati Metro station, D walls are used as soil retaining structures but in the construction of the underground tunnel which connects Michael Nagar to Birati, sheet piles are used.



**Fig: Z-Type Steel Sheet Piles at Birati Metro site**

#### 1. Introduction to Sheet Pile Installation

Sheet piling is one of the most widely used temporary retaining systems in deep excavation projects. It consists of continuous steel sections with specially designed interlocking edges that are driven vertically into the ground to form a rigid retaining wall.

At the Birati Underground Metro Project, sheet piles were installed before excavation to retain the surrounding soil, control groundwater ingress, and ensure the safety of nearby roads, utilities, and structures.

The installation was carried out using a vibratory hammer (vibro hammer) suspended from a crawler crane.

## **2. Purpose of Sheet Pile Installation**

The primary objectives of installing sheet piles at the project site were:

- \* To retain surrounding soil during excavation.
- \* To prevent collapse of excavation faces.
- \* To minimize ground settlement.
- \* To control groundwater seepage.
- \* To protect adjacent buildings and underground utilities.
- \* To provide a safe working environment.
- \* To enable construction in limited urban spaces.
- \* To reduce excavation-related risks.

## **3. Engineering Principle**

The installed sheet pile wall acts as a cantilever retaining structure embedded into the ground.

It resists:

- \* Active earth pressure
- \* Passive earth pressure
- \* Hydrostatic pressure
- \* Surcharge loads from nearby roads and buildings

The embedded portion develops passive resistance while the exposed portion supports lateral soil pressure, maintaining excavation stability.

## **4. Site Specifications (Observed During Site Visit)**

| <b>Parameter</b>           | <b>Value</b>                     |
|----------------------------|----------------------------------|
| <b>Project</b>             | Birati Underground Metro Project |
| <b>Stretch</b>             | Birati – Michael Nagar           |
| <b>Contractor</b>          | CEM India                        |
| <b>Sheet Pile Material</b> | Structural Steel                 |

| Parameter                | Value                                                                                           |
|--------------------------|-------------------------------------------------------------------------------------------------|
| Steel Grade              | Fe 350                                                                                          |
| Thickness                | 12 mm                                                                                           |
| Length of One Sheet Pile | 12 m                                                                                            |
| Combined Length Used     | 24 m (Two 12 m sheet piles welded together)                                                     |
| Weight of One Sheet Pile | Approximately 2.5 tonnes                                                                        |
| Weight of Vibro Hammer   | Approximately 8 tonnes                                                                          |
| Installation Method      | Vibratory Driving                                                                               |
| Lifting Equipment        | Hydraulic Crawler Crane                                                                         |
| Power Supply             | Diesel Generator Set                                                                            |
| Pile Driving Equipment   | Vibro Hammer with Hydraulic Clamp                                                               |
| Retaining System         | Temporary Steel Sheet Pile Wall                                                                 |
| Application              | Excavation Support for Underground Metro Station Construction                                   |
| Primary Purpose          | Soil Retention and Groundwater Control                                                          |
| Installation Sequence    | Survey → Lifting → Positioning → Clamping → Vibratory Driving → Interlocking → Final Inspection |
| Soil Condition           | Soft Alluvial Soil (Observed at Site)                                                           |
| Construction Method      | Temporary Earth Retaining System Using Vibratory Driven Steel Sheet Piles                       |

## 5. Sheet Pile Specifications

The project utilized high-strength steel sheet piles having the following characteristics:

**Steel Grade:**

The sheet piles were manufactured using Fe 350 structural steel, providing high yield strength and sufficient ductility to resist bending during installation and service.

**Thickness:**

Each sheet pile had a thickness of approximately 12 mm, ensuring adequate structural stiffness against lateral earth pressure.

**Length:**

Each individual sheet pile measured 12 metres.

For deeper excavations, two 12-metre sheet piles were joined together, forming a total length of approximately 24 metres.

The extension was achieved through welding before installation.

**6. Equipment Used****Vibro Hammer**

The primary equipment used for installation was a Vibratory Hammer.

Approximate weight: 8 tonnes

**Working Principle:**

1. The vibratory hammer grips the top of the sheet pile using hydraulic clamps.
2. Inside the hammer, rotating eccentric masses generate high-frequency vertical vibrations.
3. These vibrations reduce friction between the steel pile and surrounding soil.
4. As soil resistance decreases, the sheet pile penetrates into the ground under:

- \* Self-weight
- \* Weight of vibro hammer
- \* Additional downward force provided by the crane

Unlike impact hammers, no repeated striking occurs.

The process is smooth, faster, and produces comparatively less noise.

**7. Installation Procedure Observed at Site****Step 1 – Survey and Setting Out**

The pile alignment was marked according to the approved construction drawings.

Guide frames were installed to maintain vertical alignment.

**Step 2 – Lifting**

Each steel sheet pile weighing approximately 2.5 tonnes was lifted using a crawler crane.

The pile was positioned carefully at the predetermined location.

### **Step 3 – Connection with Vibro Hammer**

The vibro hammer hydraulically clamped the top portion of the sheet pile.

Proper gripping was checked before commencement.

### **Step 4 – Alignment**

Survey engineers checked:

- \* Verticality
- \* Orientation
- \* Position
- \* Interlocking alignment

### **Step 5 – Vibratory Driving**

The vibratory hammer generated high-frequency vibrations.

These vibrations temporarily reduced soil resistance.

The combined weight of:

- \* Vibro hammer (8 tonnes)
- \* Sheet pile (2.5 tonnes)

helped the sheet pile gradually penetrate the soil.

Driving continued until the required embedment depth was achieved.

### **Step 6 – Interlocking**

Each new sheet pile was carefully interlocked with the previously installed pile.

This formed a continuous retaining wall with minimal soil leakage.

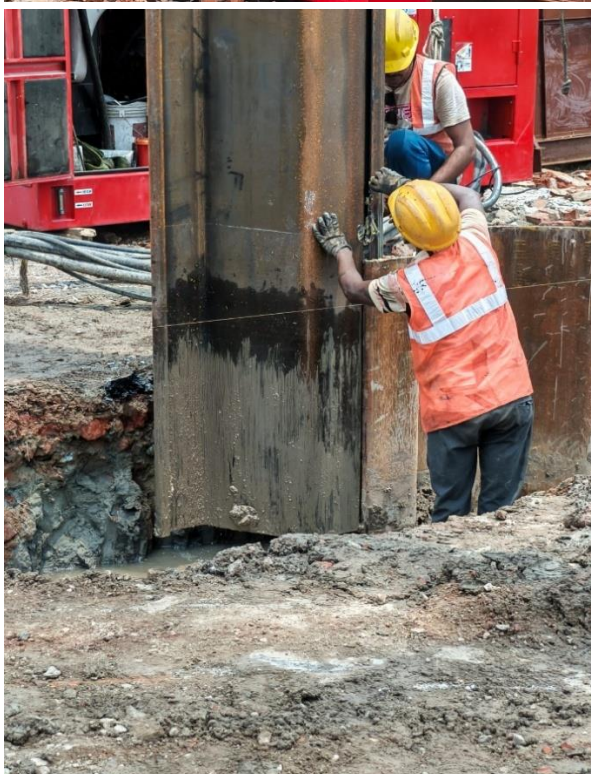
### **Step 7 – Final Inspection**

After installation, engineers checked:

- \* Final depth
- \* Verticality
- \* Interlock condition
- \* Alignment
- \* Overall stability



Fig: Installing the Sheet pile using Vibrometer At the site





**Fig: Lifting the Sheet pile by crane and some manual adjustment by labourer**

## **8. Why Vibratory Installation Was Preferred**

The vibro hammer offered several advantages:

- \* Faster installation.
- \* Lower construction time.
- \* Lower noise than impact hammers.
- \* Reduced vibration to surrounding structures.
- \* Better suitability for soft alluvial soils.
- \* Reusable equipment.
- \* Less damage to sheet piles.

## **9. Engineering Considerations**

Before selecting sheet piles, engineers considered:

- \* Soil investigation reports.
- \* Borehole data.
- \* Groundwater level.
- \* Excavation depth.
- \* Nearby buildings.

- \* Traffic loads.
- \* Utility lines.
- \* Required factor of safety.
- \* Earth pressure calculations.

## **10. Soil Behaviour During Installation**

When vibration is applied:

- \* Soil particles temporarily rearrange.
- \* Friction between soil and steel decreases.
- \* Penetration resistance reduces.
- \* The sheet pile sinks smoothly.
- \* After vibration stops, soil reconsolidates around the pile.

This mechanism makes vibratory driving highly effective in sandy and soft cohesive soils.

## **11. Quality Control**

During installation, engineers monitored:

- \* Verticality of sheet piles.
- \* Driving depth.
- \* Alignment.
- \* Interlocking quality.
- \* Welded joints (for 24 m piles).
- \* Surface damage.
- \* Vibration levels.

## **12. Safety Measures**

The following precautions were observed:

- \* PPE for all workers.
- \* Crane inspection before lifting.
- \* Restricted working zone.
- \* Safe lifting practices.
- \* Communication between crane operator and supervisor.
- \* Continuous monitoring during driving.

### **13. Advantages of Sheet Piling at the Project**

- \* Rapid construction.
- \* High structural strength.
- \* Reusable steel sections.
- \* Effective groundwater control.
- \* Minimal excavation width.
- \* Excellent support in soft soil.
- \* Safe excavation in urban areas.
- \* Reduced risk of collapse.

### **14. Challenges**

Possible construction challenges include:

- \* Driving through dense soil layers.
- \* Obstruction from buried utilities.
- \* Misalignment during installation.
- \* Interlock damage.
- \* Excessive vibration near sensitive structures.
- \* Groundwater seepage.

### **15. Practical Learning from the Site Visit**

The site visit provided valuable exposure to the practical implementation of temporary retaining systems used in underground metro construction. Observing the installation process demonstrated how theoretical concepts such as earth pressure, soil–structure interaction, and vibratory pile driving are applied in real engineering practice.

A notable observation was the use of an 8-tonne vibratory hammer to install 12 mm thick Fe 350 steel sheet piles weighing approximately 2.5 tonnes each. For greater excavation depths, two 12-metre sheet piles were welded together to form a 24-metre pile, ensuring sufficient embedment and stability. The installation process highlighted the importance of accurate alignment, continuous interlocking, and strict quality control in forming an effective retaining wall.

The visit reinforced the critical role of sheet pile systems in ensuring excavation safety, controlling groundwater, minimizing ground movement, and protecting nearby infrastructure during the construction of underground metro stations

## Concrete Batching Plant

### 1. Introduction to Concrete Batching Plants

A concrete batching plant (or batch plant) is a production system designed to combine various ingredients—including water, air, admixtures, sand, aggregate (rocks, gravel), and cement—to form concrete. These plants are essential for construction projects requiring high-quality, durable materials for infrastructure like bridges, highways, and buildings.



### 2. Classification of Batching Plants

Batching plants are categorized based on their mixing method and mobility:

(i) Based on Mixing Method :

- **Dry Mix (Transit Mix) Plants:** These plants weigh the sand, gravel, and cement, then discharge them into a **mixer truck**. Water is added either in route or at the job site, and the mixing occurs entirely within the truck's drum. This method is ideal for remote locations or projects with long-distance transportation needs.

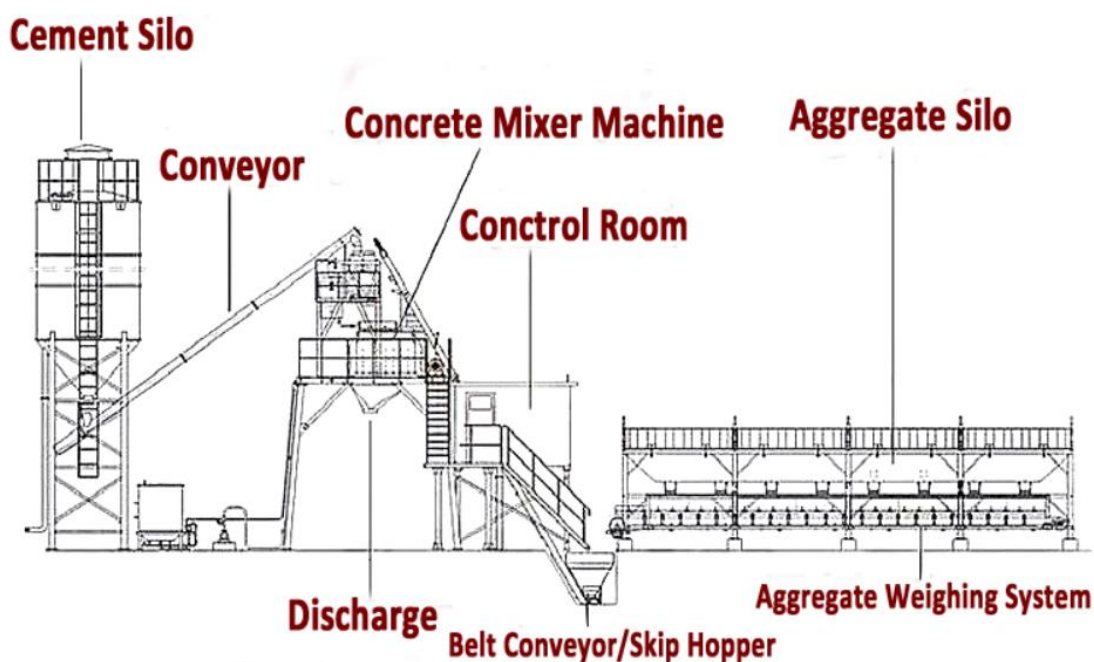
- **Wet Mix (Central Mix) Plants:** All ingredients, including water, are mixed in a **central mixer** at the plant before being loaded into a truck. This ensures greater consistency and strength reliability, making it the preferred choice for high-strength structural projects.

At our site, we use Wet Mix Plants.

## (ii) Based on Mobility :

- **Stationary Plants:** Stationary concrete plant is designed to produce high-quality concrete. It has the advantages of large output, high efficiency, high stability and high specification. Stationary concrete batching plants are reliable and flexible, easy to maintain and have a low failure rate. They are widely used in various projects such as roads and bridges, ports, tunnels, dams and buildings.
- **Mobile Plants:** The mobile batch plant, also known as a portable concrete plant, is a very productive, reliable and cost-effective piece of equipment to produce batches of concrete. It allows the user to batch concrete at almost any location, then move to another location and batch concrete. Portable plants are the best choice for temporary site projects or even stationary locations where the equipment height is a factor or the required production rate is lower.

## 3. Key Components and Their Functions



The efficiency of a plant relies on several integrated parts:

1. **Aggregate Silo & Bins:** Bins store different sizes of sand and stone. Materials are typically moved from storage piles to these bins using a **front-end loader**.
2. **Cement Silos:** Large, enclosed tanks used to store bulk cement and cementitious materials like fly ash.
3. **Weigh Hoppers:** Independent systems used to precisely weigh aggregates, cement, and water before they enter the mixer.
4. **The Mixer:** Often called the "heart" of the plant. Common types include:
  - **Twin-Shaft Mixers:** Use high horsepower for even, homogeneous mixtures.
  - **Planetary Mixers:** Achieve "dead-spot-free" mixing through complex trajectories, ideal for specialized materials like ultra-high-performance concrete.
5. **Control Panel/Automation System:** A PLC-based system that monitors ingredient measurements, saves recipes, and ensures the concrete meets specific design strengths.

#### 4. Working Principle

The batching process follows a precise sequence:

1. **Feeding:** Aggregates are fed into individual bins.
2. **Weighing:** Each ingredient is weighed according to a pre-set recipe in the control system.
3. **Transfer:** A conveyor belt or skip hoist moves the weighed aggregates to the mixing unit.
4. **Mixing:** For wet plants, the central mixer blends the materials uniformly.
5. **Discharge:** The final mix is loaded into a transit mixer or dump truck for delivery.

#### 5. Calibration and Maintenance

In modern construction, **calibration** is not just a routine—it is a safeguard for quality. It ensures that weighing elements are accurate, which directly impacts the water-cement ratio and the final concrete strength.

- **Calibration Frequency:** Typically performed every **3–6 months**, or after major maintenance.
- **Acceptable Tolerances:** Generally, cement and water must be within **±1%** accuracy, while aggregates should be within **±2%**.
- **Daily Maintenance:** Includes washing out the central mixer drum and ensuring all safety switches and emergency buttons function properly.

### Step-by-Step Calibration Procedure:

The standard workflow for calibrating a batching plant involves several stages:

- **Pre-Check and Safety:** The plant is switched to safe mode with power isolation (lock-out/tag-out). Operators inspect load-cell cables, junction boxes, and remove any material buildup or mechanical friction that could distort readings.
- **Zero and Dead Load Check:** The system is calibrated to **zero** when completely empty. A stable zero reading must remain constant for 2–3 minutes before proceeding.
- **Incremental Loading:** Certified test weights are added at typical test points, such as **5%, 20%, 40%, 60%, 80%, and 100%** of full capacity.
- **sReading:** Measurements are recorded while both **increasing and decreasing** the load to detect **hysteresis** (the difference in readings when loading vs. unloading).
- **Adjustment (Span/Linearization):** The span is adjusted to align the displayed weight with the actual test weight. Multi-point linearization may be used for higher precision.
- **Verification and Documentation:** All points are re-measured after adjustment to confirm accuracy. Finally, raw data and deviation charts are compiled into a report, and each weighing point is labeled with the calibration and next due date.

## Ready Mixed Concrete (RMC) Plants

RMC plants are centralised facilities where concrete is manufactured according to a set recipe and then delivered to construction sites in a "ready-to-use" state. Unlike on-site batching, RMC plants serve multiple customers and projects simultaneously, often operating within urban areas to minimize travel time.

### 1. Advanced Requirements and Standards

Because they supply critical infrastructure, RMC plants have higher requirements for **durability, reliability, safety, and environmental protection** compared to temporary on-site plants. They are essential for city projects where space for raw material storage is limited.

### 2. Digital Control and Automation

Modern industrial standards (such as IS 4925) mandate that RMC plants be highly automated to ensure consistency:

- **System Equipment:** Plants must be equipped with a computer system capable of storing a minimum of **15 different recipes**, alongside a monitor and printer for real-time tracking.
- **Traceability:** Systems often use **moisture probes** to measure the water content in sand and aggregates as they enter the batcher, automatically adjusting the mix to maintain the correct water-cement ratio.

### 3. Mandatory Delivery Documentation

Quality control is finalized through a **delivery slip** that must accompany every transit mixer. This document is vital for project audits and must include:

- **Supplier & Recipe Info:** Plant name, logo, recipe number, and the specific ordered quantity in m<sup>3</sup>.
- **Material Accuracy:** A weighing chart for each component and the **moisture percentage** of materials at the time of batching.
- **Timing & Logistics:** Precise start and finish times for the batch, the transit mixer number, and the truck number.

Some photographs that we have taken during our plant visit:

**L.K. BUILDERS**

**Report Title:** Batch Sheet

**Plant Sl.No:** 222

|                                     |                              |                           |  |
|-------------------------------------|------------------------------|---------------------------|--|
| Batch Date : 10-Jun-26              | Batch Start Time : 03:24:03  | Batch End Time : 03:30:09 |  |
| Batch Number : 26715.00             | Recipe Code : CEM M40 D WALL | Ordered Qty : 150.00      |  |
| Batcher Name : Stetter              | Recipe Name : CEM M40 D WALL | Production Qty : 6.50     |  |
| Order Number : CEM M40 D WALL       | Truck Number : WB25K 9803    | Adj/Manual Qty : 0.00     |  |
| Customer : CEMINDIA PROJECTS LIMITE | Truck Driver : SOFIKUL ISLAM | With This Load : 6.50     |  |
| Site : EPC-BIRATI                   | Batch Size : 1.08            | Mixer Capacity : 1.25     |  |

| Aggregate |      |      |      | Cement |      |      |      | Water/Ice |      |      | Admixture |      |         |         | Silica  |         |        |
|-----------|------|------|------|--------|------|------|------|-----------|------|------|-----------|------|---------|---------|---------|---------|--------|
| 10MM SAND | 40MM | 20MM | NILL | NILL   | Cem1 | Cem2 | Cem3 | NILL      | NILL | Wtr1 | Wtr2      | Wtr3 | Admix 1 | Admix 2 | Admix 3 | Admix 4 | Silica |
| 516       | 738  | 0    | 631  | 0      | 0    | 450  | 0    | 0         | 0    | 162  | 0         | 0    | 3.15    | 0.00    | 0.00    | 0.00    | 0      |

**Mass of Recipe Targets in Kgs.**      2500.15

Target and Actual Value with moisture correction/absorption in % and other Corrections in Kgs.

|      |      |      |      |      |      |   |     |   |   |   |     |   |   |      |      |      |      |   |
|------|------|------|------|------|------|---|-----|---|---|---|-----|---|---|------|------|------|------|---|
| 559  | 799  | 0    | 684  | 0    | 0    | 0 | 487 | 0 | 0 | 0 | 175 | 0 | 0 | 3.41 | 0.00 | 0.00 | 0.00 | 0 |
| 545  | 806  | 0    | 678  | 0    | 0    | 0 | 481 | 0 | 0 | 0 | 176 | 0 | 0 | 3.41 | 0.00 | 0.00 | 0.00 | 0 |
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0 | 0   | 0 | 0 | 0 | 0   | 0 |   | 0.00 | 0.00 | 0.00 | 0.00 | 0 |
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |   |     |   |   |   |     |   |   |      |      |      |      |   |
| 559  | 799  | 0    | 684  | 0    | 0    | 0 | 487 | 0 | 0 | 0 | 175 | 0 | 0 | 3.41 | 0.00 | 0.00 | 0.00 | 0 |
| 558  | 807  | 0    | 682  | 0    | 0    | 0 | 481 | 0 | 0 | 0 | 177 | 0 | 0 | 3.41 | 0.00 | 0.00 | 0.00 | 0 |
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0 | 0   | 0 | 0 | 0 | 0   | 0 |   | 0.00 | 0.00 | 0.00 | 0.00 | 0 |
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |   |     |   |   |   |     |   |   |      |      |      |      |   |
| 559  | 799  | 0    | 684  | 0    | 0    | 0 | 487 | 0 | 0 | 0 | 175 | 0 | 0 | 3.41 | 0.00 | 0.00 | 0.00 | 0 |
| 555  | 804  | 0    | 674  | 0    | 0    | 0 | 487 | 0 | 0 | 0 | 176 | 0 | 0 | 3.31 | 0.00 | 0.00 | 0.00 | 0 |
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0 | 0   | 0 | 0 | 0 | 0   | 0 |   | 0.00 | 0.00 | 0.00 | 0.00 | 0 |
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |   |     |   |   |   |     |   |   |      |      |      |      |   |
| 559  | 799  | 0    | 684  | 0    | 0    | 0 | 487 | 0 | 0 | 0 | 175 | 0 | 0 | 3.41 | 0.00 | 0.00 | 0.00 | 0 |
| 565  | 799  | 0    | 684  | 0    | 0    | 0 | 489 | 0 | 0 | 0 | 175 | 0 | 0 | 3.44 | 0.00 | 0.00 | 0.00 | 0 |
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0 | 0   | 0 | 0 | 0 | 0   | 0 |   | 0.00 | 0.00 | 0.00 | 0.00 | 0 |
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |   |     |   |   |   |     |   |   |      |      |      |      |   |
| 559  | 799  | 0    | 684  | 0    | 0    | 0 | 487 | 0 | 0 | 0 | 175 | 0 | 0 | 3.41 | 0.00 | 0.00 | 0.00 | 0 |
| 556  | 803  | 0    | 684  | 0    | 0    | 0 | 489 | 0 | 0 | 0 | 178 | 0 | 0 | 3.44 | 0.00 | 0.00 | 0.00 | 0 |
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0 | 0   | 0 | 0 | 0 | 0   | 0 |   | 0.00 | 0.00 | 0.00 | 0.00 | 0 |
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |   |     |   |   |   |     |   |   |      |      |      |      |   |
| 559  | 799  | 0    | 684  | 0    | 0    | 0 | 487 | 0 | 0 | 0 | 175 | 0 | 0 | 3.41 | 0.00 | 0.00 | 0.00 | 0 |
| 568  | 774  | 0    | 686  | 0    | 0    | 0 | 494 | 0 | 0 | 0 | 177 | 0 | 0 | 3.45 | 0.00 | 0.00 | 0.00 | 0 |
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0 | 0   | 0 | 0 | 0 | 0   | 0 |   | 0.00 | 0.00 | 0.00 | 0.00 | 0 |
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |   |     |   |   |   |     |   |   |      |      |      |      |   |

**Total Set Weight in Kgs.**

|      |      |   |      |   |   |   |      |   |   |   |      |   |   |       |      |      |      |   |
|------|------|---|------|---|---|---|------|---|---|---|------|---|---|-------|------|------|------|---|
| 3354 | 4794 | 0 | 4104 | 0 | 0 | 0 | 2922 | 0 | 0 | 0 | 1050 | 0 | 0 | 20.46 | 0.00 | 0.00 | 0.00 | 0 |
|------|------|---|------|---|---|---|------|---|---|---|------|---|---|-------|------|------|------|---|

**Mass of Total Set Weight in Kgs.**      16244.48

**Total Actual Weight in Kgs.**

|      |      |   |      |   |   |   |      |   |   |   |      |   |   |       |      |      |      |   |
|------|------|---|------|---|---|---|------|---|---|---|------|---|---|-------|------|------|------|---|
| 2345 | 4793 | 0 | 4088 | 0 | 0 | 0 | 2921 | 0 | 0 | 0 | 1059 | 0 | 0 | 20.46 | 0.00 | 0.00 | 0.00 | 0 |
|------|------|---|------|---|---|---|------|---|---|---|------|---|---|-------|------|------|------|---|

**Mass of Total Actual Weight in Kgs.**      16226.48

**Difference in Percentage**

|       |       |   |       |   |   |   |       |   |   |   |      |   |   |      |   |   |   |   |
|-------|-------|---|-------|---|---|---|-------|---|---|---|------|---|---|------|---|---|---|---|
| -0.27 | -0.02 | 0 | -0.39 | 0 | 0 | 0 | -0.03 | 0 | 0 | 0 | 0.86 | 0 | 0 | 0.00 | 0 | 0 | 0 | 0 |
|-------|-------|---|-------|---|---|---|-------|---|---|---|------|---|---|------|---|---|---|---|



## Admixtures

In concrete production, liquid chemical admixtures and specialized additives like crystalline waterproofing admixtures are essential for enhancing the performance, durability, and workability of the mix.

### 1. Liquid Chemical Admixtures:

Chemical admixtures are ingredients added to concrete (other than water, aggregates, and cement) to achieve specific property or performance enhancements. These admixtures are added to concrete during batching to improve workability, control setting time, reduce water demand and enhance the strength and durability of concrete. They are dosed according to the approved concrete mix design.

In a batching plant, these are typically stored in separate tanks in liquid form.



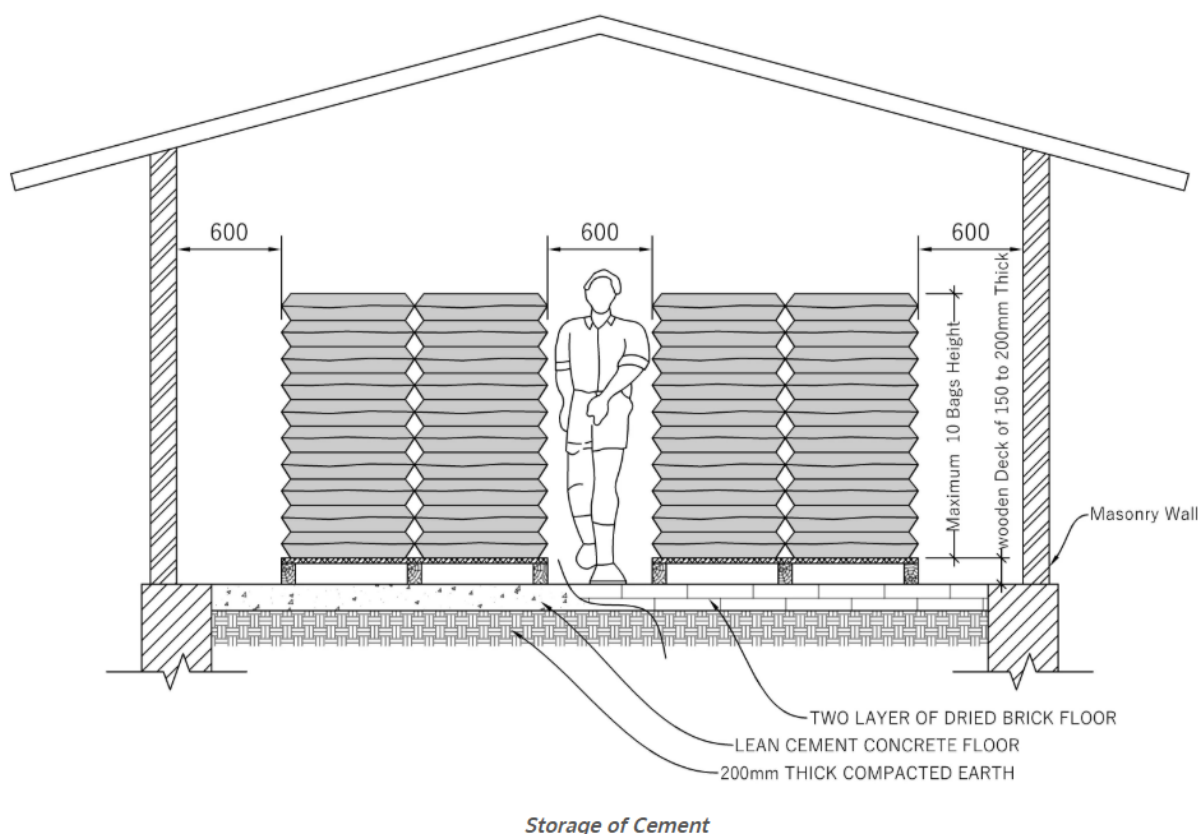
### 2. Crystalline Waterproofing Admixture:

The crystalline waterproofing admixture is added to concrete to reduce water permeability and enhance durability. It reacts with water and cement to form insoluble crystals that block pores and micro-cracks and makes the concrete more resistant to water ingress. It is widely used in underground structures, tunnels and basements.



### Storage and Stacking of Cement Bags (at Site) :

To prevent premature hydration, maintain structural integrity, and mitigate the risk of lumping under compressive pressure, cement bags must be stored in strict compliance with standard warehousing protocols (such as IS 4082).



*Storage of Cement*

#### 1. Stack Dimensions and Structural Stability

- **Maximum Stack Height:** Stacks must not exceed a maximum height of **10 bags** for short-term storage. For long-term storage exceeding two months, the height must be restricted to **7 bags** to prevent compressive damage.
- **Stack Width:** The total width of any single stack must not exceed **4 bags** or **3.0 metres**.
- **Interlocking Pattern:** To ensure structural stability and prevent toppling, bags must be laid in an alternating **header-and-stretcher (crosswise and lengthwise) pattern** for every successive layer.



- **Stack Density:** Bags within a stack must be placed as close together as possible to minimize air circulation and reduce exposure to atmospheric moisture.

## 2. Spatial Clearances and Weatherproofing

- **Plinth Clearance:** Cement must never be stored directly on the floor. Stacks must be elevated on raised wooden pallets, planks, or timber battens providing a minimum clear height of **150 mm to 200 mm** from the ground.
- **Wall Clearance:** A minimum unobstructed distance of **600 mm** must be maintained between the cement stacks and the warehouse walls to prevent moisture migration.
- **Environmental Control:** The storage facility must be completely leak-proof, dry, and feature minimal windows. For open-site or monsoon storage, stacks must be entirely enveloped in heavy-duty waterproof membranes (minimum 700-gauge polythene sheets), securely anchored at the base.

Below is a photograph of a storage room near the batching plant.



## **Quality Assurance and Quality Control**

### **Based on ISO 9001:2015 Quality Management System (QMS)**

#### **1. Introduction**

Quality is a critical aspect of every organization as it ensures customer satisfaction, regulatory compliance, and continual improvement. The ISO 9001:2015 Quality Management System (QMS) is an internationally recognized standard that provides a framework for organizations to consistently deliver products and services meeting customer and statutory requirements.

Quality Management consists of two important components:

- Quality Assurance (QA): Focuses on preventing defects by improving processes.
- Quality Control (QC): Focuses on detecting defects through inspection and testing.

#### **2. Concept of Quality Management**

Quality management refers to the coordinated activities that direct and control an organization with regard to quality. It involves establishing quality objectives, defining responsibilities, implementing standardized procedures, monitoring performance, measuring outcomes, and continually improving organizational processes. The ultimate objective of quality management is to satisfy customers by consistently providing products and services that meet their expectations while simultaneously improving operational efficiency and reducing unnecessary costs.

A Quality Management System integrates all organizational activities into a structured framework that ensures consistency in planning, execution, monitoring, evaluation, and improvement. Rather than relying solely on final inspections, modern quality management emphasizes preventing problems through systematic planning, employee competence, effective communication, documented procedures, and continual monitoring of processes.

#### **3. Quality Assurance (QA)**

##### **Definition**

Quality Assurance is a planned and systematic approach implemented within the Quality Management System to provide confidence that products or services will consistently meet specified quality requirements.

## **Objectives**

The objectives of Quality Assurance are broad and focus primarily on improving organizational processes. These objectives include preventing defects before they occur, ensuring compliance with customer requirements and international standards, improving consistency in operations, enhancing employee competence through training, establishing standardized procedures, reducing process variation, minimizing waste and rework, increasing operational efficiency, strengthening customer confidence, and supporting continual improvement within the organization. By achieving these objectives, Quality Assurance contributes to the overall effectiveness of the Quality Management System and enhances organizational competitiveness.

## **Key QA Activities**

- Preparation of Quality Management Plan.
- Documentation and control of procedures.
- Employee training and competency development.
- Internal audits.
- Risk-based thinking.
- Corrective and preventive actions.
- Management review meetings.

## **Benefits**

- Consistent product quality.
- Reduced rework and waste.
- Improved operational efficiency.
- Better customer confidence.
- Compliance with legal and regulatory requirements.

## **4. Quality Control (QC)**

### **Definition**

Quality Control is the operational technique used to verify that products or services conform to specified quality requirements through inspection, measurement, and testing.

### **Objectives**

The primary objectives of Quality Control are to verify compliance with technical specifications, detect defects before products reach customers, ensure product reliability and safety, reduce rejection rates, minimize customer complaints, maintain consistency in product

quality, provide accurate inspection records, and support continuous process improvement through feedback obtained from inspections and testing. Quality Control ultimately serves as the final verification mechanism that confirms whether quality objectives have been achieved.

### **Key QC Activities**

- Incoming material inspection.
- In-process inspection.
- Final product inspection.
- Laboratory testing.
- Calibration of measuring equipment.
- Recording inspection results.
- Non-conformance reporting.

### **Benefits**

- Defect-free products.
- Reduced rejection rates.
- Increased customer satisfaction.
- Improved product reliability.
- Better decision-making using inspection data.

## **5. Difference Between Quality Assurance and Quality Control**

Although Quality Assurance and Quality Control work together within a Quality Management System, they differ significantly in their purpose and approach. Quality Assurance focuses on improving the processes used to produce products or deliver services, whereas Quality Control concentrates on evaluating the finished product through inspection and testing. QA emphasizes prevention of defects by developing effective systems and standardized procedures, while QC identifies and corrects defects after they occur. QA activities include quality planning, documentation, employee training, internal auditing, and process improvement, whereas QC activities include inspections, laboratory testing, measurements, calibration, and acceptance or rejection of products. Together, they ensure both process effectiveness and product conformity.

## **6. ISO 9001:2015 Quality Management System**

ISO 9001:2015 provides a structured framework for organizations to manage quality effectively and achieve continual improvement. The standard follows the **Plan-Do-Check-Act (PDCA)** methodology, which promotes systematic planning, implementation, monitoring, evaluation, and improvement of organizational processes. Additionally, ISO 9001:2015 introduces **risk-based thinking**, encouraging organizations to identify potential risks and

opportunities during planning so that preventive measures can be implemented before problems occur.

The standard emphasizes customer satisfaction as the primary objective of the Quality Management System. Organizations are required to understand customer expectations, monitor customer satisfaction, evaluate organizational performance, and continually improve their processes to meet changing customer needs. ISO 9001:2015 also promotes evidence-based decision making by encouraging organizations to analyze data, monitor performance indicators, conduct internal audits, and implement improvements based on factual information rather than assumptions.

## **7. Clauses of ISO 9001:2015 Quality Management System**

The **ISO 9001:2015 Quality Management System (QMS)** consists of **10 clauses**, of which **Clauses 1–3 are introductory** and **Clauses 4–10 contain the mandatory requirements** for implementing and maintaining a Quality Management System.

### **Clause 1: Scope**

This clause defines the scope and purpose of ISO 9001:2015. It specifies the requirements for a Quality Management System that enables an organization to consistently provide products and services meeting customer, statutory, and regulatory requirements while enhancing customer satisfaction through the effective implementation of the QMS. The standard is applicable to organizations of all types, sizes, and sectors, regardless of the nature of their products or services.

### **Clause 2: Normative References**

This clause identifies the documents that are indispensable for the application of ISO 9001:2015. It refers to **ISO 9000:2015 – Quality Management Systems: Fundamentals and Vocabulary**, which provides the principles, concepts, and terminology used throughout the standard. Understanding these terms ensures consistent interpretation and implementation of the QMS requirements.

### **Clause 3: Terms and Definitions**

Clause 3 provides the key terms and definitions used in ISO 9001:2015. These definitions help organizations understand concepts such as quality, customer satisfaction, risk, process, nonconformity, corrective action, documented information, and continual improvement. A common understanding of these terms is essential for effective communication and implementation of the Quality Management System.

### **Clause 4: Context of the Organization**

This clause requires organizations to understand the internal and external factors that can influence their ability to achieve intended quality outcomes. Organizations must identify the needs and expectations of interested parties such as customers, suppliers, regulatory authorities, employees, and shareholders. They must define the scope of the Quality Management System and establish the processes necessary for its implementation.

Understanding the organizational context ensures that the QMS is aligned with strategic objectives and business operations.

### **Clause 5: Leadership**

Leadership plays a vital role in the success of the Quality Management System. This clause requires top management to demonstrate commitment by establishing a quality policy, defining quality objectives, allocating resources, promoting customer focus, and ensuring that responsibilities and authorities are clearly assigned. Top management must actively participate in maintaining the effectiveness of the QMS and encourage a culture of continual improvement throughout the organization.

### **Clause 6: Planning**

Clause 6 focuses on planning actions to address risks and opportunities that may affect the organization's ability to achieve quality objectives. Organizations must identify potential risks, evaluate opportunities for improvement, establish measurable quality objectives, and prepare plans to achieve those objectives. Planning also includes managing organizational changes in a controlled manner to ensure the continued effectiveness of the Quality Management System.

### **Clause 7: Support**

This clause deals with the resources and support required for the effective implementation of the Quality Management System. Organizations must provide adequate human resources, infrastructure, work environment, monitoring and measuring equipment, organizational knowledge, and financial resources. Employees must be competent through appropriate education, training, and experience. The clause also emphasizes awareness, communication, and control of documented information such as procedures, records, manuals, and work instructions. Proper support ensures that all quality-related activities are performed effectively and consistently.

### **Clause 8: Operation**

Clause 8 addresses the operational processes required to deliver products and services that meet customer requirements. Organizations must plan and control operational activities, review customer requirements, design and develop products where applicable, control externally provided products and services, manage production and service provision, identify and preserve products, protect customer property, control changes, and verify that outputs conform to specified requirements before delivery. Any non-conforming outputs must be identified and controlled to prevent unintended use or delivery. This clause forms the operational core of the Quality Management System.

### **Clause 9: Performance Evaluation**

Organizations must regularly monitor, measure, analyze, and evaluate the effectiveness of their Quality Management System. This clause requires the organization to assess customer satisfaction, monitor process performance, analyze quality data, conduct internal audits, and perform management reviews. Performance evaluation helps identify strengths, weaknesses,

and opportunities for improvement, ensuring that the Quality Management System continues to achieve its intended results.

### **Clause 10: Improvement**

The final clause emphasizes continual improvement of the Quality Management System. Organizations must identify nonconformities, investigate their root causes, implement corrective actions, and verify the effectiveness of those actions. Lessons learned from audits, customer feedback, process performance, and quality data should be used to improve products, services, and organizational processes. Continual improvement enhances customer satisfaction, increases operational efficiency, and strengthens the organization's ability to meet future challenges.

## **8. PDCA Cycle (Plan–Do–Check–Act)**

The **PDCA (Plan–Do–Check–Act) Cycle**, also known as the **Deming Cycle** or **Shewhart Cycle**, is a systematic and continuous improvement methodology that forms the foundation of the **ISO 9001:2015 Quality Management System (QMS)**. It helps organizations plan their activities, implement processes effectively, evaluate performance, and take corrective actions to achieve continual improvement. By following the PDCA cycle, organizations can improve product quality, enhance customer satisfaction, reduce process variations, and ensure that quality objectives are consistently achieved.

### **Plan**

The **Plan** phase is the first stage of the PDCA cycle and involves identifying customer requirements, understanding organizational objectives, and establishing the processes needed to achieve the desired results. During this stage, organizations analyze internal and external issues, identify potential risks and opportunities, define quality objectives, allocate resources, and prepare detailed plans for implementation. Quality policies, inspection and test plans, standard operating procedures, and work instructions are developed during this phase. Proper planning ensures that activities are carried out systematically and helps prevent quality problems before they occur.

### **Do**

The **Do** phase involves implementing the plans developed during the planning stage. In this phase, the organization carries out production or service activities according to established procedures and quality plans. Employees perform their assigned tasks, machinery and equipment are operated under controlled conditions, and quality assurance measures are applied throughout the process. Personnel receive appropriate training, resources are utilized efficiently, and all activities are documented. The objective of this stage is to execute the planned processes consistently while ensuring compliance with quality requirements.

### **Check**

The **Check** phase focuses on monitoring, measuring, and evaluating the performance of processes and products. Organizations collect data through inspections, testing, internal audits, customer feedback, and performance measurements to determine whether quality objectives have been achieved. Any deviations, defects, or non-conformities are identified and analyzed to determine their causes. This phase provides objective evidence of process effectiveness and helps management evaluate whether the Quality Management System is functioning as intended.

### **Act**

The **Act** phase is the final stage of the PDCA cycle and emphasizes continual improvement. Based on the findings from the Check phase, organizations take corrective actions to eliminate the root causes of identified non-conformities and implement preventive measures to avoid their recurrence. Management reviews process performance, updates quality objectives and procedures where necessary, and introduces improvements to enhance efficiency, productivity, and customer satisfaction. The lessons learned during this phase become the basis for the next planning cycle, making quality improvement a continuous process.

### **Importance of the PDCA Cycle**

The PDCA cycle provides a structured framework for managing and improving organizational processes. It promotes a proactive approach to problem-solving by encouraging organizations to identify risks, monitor performance, and implement improvements continuously. In ISO 9001:2015, the PDCA cycle supports effective process management, enhances customer satisfaction, improves operational efficiency, reduces defects and waste, ensures compliance with statutory and regulatory requirements, and fosters a culture of continual improvement throughout the organization.

## **9. Documentation in ISO 9001:2015**

Documentation forms the backbone of the Quality Management System because it provides objective evidence that organizational activities have been performed according to established procedures. Important quality documents include the Quality Policy, Quality Objectives, Project Quality Plans, Standard Operating Procedures (SOPs), Inspection and Test Plans (ITPs), Method Statements, Work Instructions, Material Test Certificates, Inspection Reports, Calibration Certificates, Audit Reports, Corrective Action Reports, Preventive Action Reports, Non-Conformance Reports, and Management Review Records. Proper documentation ensures traceability, consistency, accountability, and compliance with both customer requirements and regulatory standards.

## **10. Benefits of ISO 9001:2015**

Organizations implementing ISO 9001:2015 experience numerous benefits, including improved customer satisfaction, enhanced product quality, standardized work procedures, reduced operational costs, lower defect rates, improved resource utilization, better risk management, increased employee involvement, stronger supplier relationships, improved regulatory compliance, enhanced organizational reputation, and greater opportunities in national and international markets. The emphasis on continual improvement enables

organizations to remain competitive while adapting to changing customer expectations and technological advancements.

## **11. Application in the Construction Industry**

In the construction industry, Quality Assurance and Quality Control play a crucial role in ensuring that projects meet structural, functional, and safety requirements. QA activities include preparing project quality plans, reviewing design documents, approving construction procedures, training personnel, and conducting internal audits. QC activities involve testing construction materials such as cement, aggregates, concrete, reinforcement steel, and soil, as well as inspecting excavation, reinforcement placement, formwork, concreting, welding, and finishing works. Continuous documentation, monitoring, calibration of equipment, and final inspections ensure compliance with engineering specifications, contractual obligations, and relevant national and international standards. Effective implementation of QA/QC practices enhances structural durability, minimizes defects, reduces project delays, improves safety, and increases client satisfaction.

## **12. Conclusion**

Quality Assurance and Quality Control are essential elements of the ISO 9001:2015 Quality Management System. QA focuses on preventing defects by establishing effective processes, while QC ensures that products and services meet specified quality requirements through inspection and testing. Together, they enhance customer satisfaction, improve organizational performance, and promote continual improvement. By implementing ISO 9001:2015, organizations can achieve consistent quality, operational excellence, and long-term business success.

## **LAB EXPERIMENTS IN QUALITY CONTROL LABORATORY**

### **CEMENT TESTING:**

To determine the quality, strength, consistency, and durability of cement, ensuring it meets the required standards for construction. Cement testing helps verify properties such as fineness, setting time, soundness, specific gravity, and compressive strength, thereby ensuring proper workability, adequate strength development, volume stability, and long-term performance of concrete structures while preventing premature failures and enhancing the safety, durability, and cost-effectiveness of the project.

### **TEST FOR CEMENT:**

#### **➤ Standard Consistency Test:**

##### **1. Aim of the Test:**

- To determine the percentage of water required to prepare a cement paste of standard consistency.

##### **2. Theory:**

- Standard consistency is the amount of water required to produce a cement paste of standard plasticity. It is determined using the Vicat apparatus and is required for conducting other cement tests such as setting time and soundness.

##### **3. Procedure (As per IS:4031 Part 4:1988):**

- Weigh 400 g of cement and prepare a cement paste by adding a measured quantity of water.
- Fill the Vicat mould completely with the paste and level the top surface.
- Place the mould under the Vicat apparatus fitted with a 10 mm diameter plunger.
- Release the plunger gently and allow it to penetrate the paste.
- Repeat the test with different water contents until the plunger penetrates 33–35 mm from the top of the mould (or 5–7 mm from the bottom).
- The corresponding water content, expressed as a percentage of the weight of cement, is reported as the standard consistency.

##### **4. Acceptance / Standards (As per IS:455-2015):**

- No specific acceptance value is prescribed.

- The standard consistency is determined experimentally and generally lies between 26% and 33% by weight of cement.

5. **Code:**

- IS:4031 (Part 4):1988 – Methods of Physical Tests for Hydraulic Cement – Determination of Consistency of Standard Cement Paste.
- IS:455:2015 – Portland Slag Cement – Specification.

➤ **Initial Setting Time:**

1. **Aim of the Test:**

- To determine the time required for cement paste to begin losing its plasticity after the addition of water.

2. **Theory:**

- Initial setting time is the time elapsed from the addition of water to cement until the paste starts losing its plasticity. It indicates the available time for mixing, transporting, placing, and compacting concrete. The test is carried out using the Vicat apparatus.

3. **Procedure (As per IS:4031 Part 5:1988):**

- Prepare a cement paste using 0.85 times the water required for standard consistency (0.85P).
- Fill the Vicat mould completely with the cement paste and level the surface.
- Place the mould under the Vicat apparatus fitted with the 1 mm square needle.
- Allow the needle to penetrate the paste at regular intervals.
- Record the time from the addition of water until the needle penetrates only 33–35 mm from the top of the mould (or  $5 \pm 0.5$  mm from the bottom).
- This elapsed time is reported as the initial setting time of cement.

4. **Acceptance / Standards (As per IS:455-2015):**

- Initial setting time shall not be less than 30 minutes.

5. **Code:**

- IS:4031 (Part 5):1988 – Methods of Physical Tests for Hydraulic Cement – Determination of Initial and Final Setting Times.
- IS:455:2015 – Portland Slag Cement – Specification.

➤ **Final Setting Time:**

**1. Aim of the Test:**

- To determine the time required for cement paste to completely lose its plasticity and attain sufficient hardness after the addition of water.

**2. Theory:**

- Final setting time is the time elapsed from the addition of water to cement until the paste has completely lost its plasticity and attained adequate rigidity. It indicates the time after which the cement paste can withstand a certain amount of pressure without appreciable deformation. The test is conducted using the Vicat apparatus.

**3. Procedure (As per IS:4031 Part 5:1996):**

- Prepare a cement paste using 0.85 times the water required for standard consistency (0.85P).
- Start the stopwatch immediately after adding water to the cement.
- Fill the Vicat mould completely with the cement paste placed on a non-porous plate and level the surface.
- Keep the mould in a moist environment except during testing.
- Replace the initial setting needle with the needle having an annular attachment.
- Lower the needle gently onto the surface of the cement paste at regular intervals.
- The cement is considered finally set when the needle makes an impression on the surface but the annular attachment fails to make any impression.
- The time elapsed from the addition of water to this stage is recorded as the final setting time and reported to the nearest 5 minutes.

**4. Acceptance / Standards (As per IS:455-2015):**

- Final setting time shall not be more than 600 minutes (10 hours).

**5. Code:**

- IS:4031 (Part 5):1996 – Methods of Physical Tests for Hydraulic Cement – Determination of Initial and Final Setting Times.
- IS:455:2015 – Portland Slag Cement – Specification.

➤ **Compressive Strength:**

**1. Aim of the Test:**

- To determine the compressive strength of standard cement mortar cubes at specified ages (3, 7 and 28 days) for assessing the quality and strength characteristics of hydraulic cement.

**2. Theory:**

- The compressive strength of cement is determined by testing 70.6 mm mortar cubes prepared with cement and standard sand in the ratio 1:3 by weight. The mortar is compacted on a standard vibration machine, cured under specified conditions, and tested in a compression testing machine. The compressive strength is calculated by dividing the maximum load at failure by the loaded area of the cube.

**3. Procedure (As per IS:4031 Part 6:1988):**

- Prepare mortar by mixing 1 part cement and 3 parts standard sand with the specified quantity of water.
- Fill 70.6 mm cube moulds in two layers and compact using a standard vibration machine.
- Keep the specimens in a moist atmosphere for  $24 \pm \frac{1}{2}$  hours.
- Demould the cubes and cure them in clean water at  $27 \pm 2^\circ\text{C}$  until the time of testing.
- Test the cubes in a compression testing machine at 3, 7 and 28 days.
- Record the maximum load at failure and calculate the compressive strength by dividing the load by the cross-sectional area of the cube.
- Report the average compressive strength of the valid test specimens.

**4. Acceptance / Standards (As per IS:455-2015):**

| Age of Test | Minimum Compressive Strength |
|-------------|------------------------------|
| 3 Days      | 16 MPa                       |
| 7 Days      | 22 MPa                       |
| 28 Days     | 33 MPa                       |

- The cement complies with IS 455:2015 if the average compressive strength is not less than the above specified values.

5. Code:

- IS 4031 (Part 6): 1988 – Methods of Physical Tests for Hydraulic Cement – Determination of Compressive Strength of Hydraulic Cement (Other than Masonry Cement).
- IS 455:2015 – Portland Slag Cement – Specification.

➤ **Fineness:**

**1. Aim of the Test:**

- To determine the fineness of cement by measuring the percentage residue retained on a 90  $\mu$ m IS sieve, which indicates the degree of grinding and uniformity of the cement.

**2. Theory:**

- Fineness of cement influences the rate of hydration, strength development, heat of hydration, and workability. It is determined by sieving a known quantity of cement through a 90  $\mu$ m IS sieve. The percentage of cement retained on the sieve is taken as the measure of fineness.

**3. Procedure (As per IS:4031 Part 1:1988):**

- Take 100 g of cement sample.
- Place the sample on a clean and dry 90  $\mu$ m IS sieve fitted with a pan.
- Sieve the cement continuously for 15 minutes using gentle circular and vertical motions.
- Break any air-set lumps gently with fingers without forcing them through the sieve.
- Weigh the residue retained on the sieve.
- Calculate the percentage residue by mass on the 90  $\mu$ m sieve and report the average of the test results.

**4. Acceptance / Standards (As per IS:455-2015):**

- The residue on the 90  $\mu$ m IS sieve shall not exceed 10% by mass.

**5. Code:**

- IS 4031 (Part 1): 1988 – Methods of Physical Tests for Hydraulic Cement – Determination of Fineness by Dry Sieving.
- IS 455:2015 – Portland Slag Cement – Specification.

## **AGGREGATE TESTING:**

To determine the physical, mechanical, and durability properties of aggregates, ensuring they are suitable for use in pavement and concrete works. Aggregate testing helps verify properties such as gradation, specific gravity, water absorption, flakiness index, elongation index, impact value, and crushing value, thereby ensuring adequate strength, proper interlocking, good load-bearing capacity, and long-term durability of pavement layers while preventing premature failures and enhancing the safety, quality, and cost-effectiveness of the project.

## **TEST FOR COARSE AGGREGATE:**

### **➤ Particle Size and Shape Test:**

#### **1. Aim of the Test:**

- To determine the particle size distribution (grading) and particle shape (flakiness index and elongation index) of coarse aggregates for assessing their suitability for concrete production.

#### **2. Theory:**

- The particle size distribution is determined by sieve analysis, which indicates the grading of the aggregate and affects workability, strength, and void content of concrete. The shape of aggregates is evaluated by determining the Flakiness Index (FI) and Elongation Index (EI) using standard gauges. Excessively flaky or elongated particles reduce the strength, durability, and workability of concrete.

#### **3. Procedure (As per IS:2386 Part 1:1963):**

- Collect a representative aggregate sample and dry it to constant mass.
- Perform sieve analysis using the specified IS sieves and determine the percentage retained on each sieve to obtain the grading curve.
- Separate aggregate into different size fractions for shape tests.
- Determine the Flakiness Index by passing particles through the appropriate thickness gauge and calculating the percentage by mass of flaky particles.
- Determine the Elongation Index by checking particles with the standard length gauge and calculating the percentage by mass of elongated particles.
- Record and report the grading, Flakiness Index, and Elongation Index.

#### **4. Acceptance / Standards (As per IS:383-2016):**

- Grading: Shall conform to the grading limits specified in IS 383:2016 for the selected nominal size of coarse aggregate and satisfy the requirements of the approved design mix.
- Flakiness Index (FI): Preferably not more than 30% for concrete works.
- Elongation Index (EI): Preferably not more than 30% for concrete works.
- Combined Flakiness and Elongation Index (FI + EI): Should not exceed 35% for aggregates used in pavement and concrete applications, unless otherwise specified in the project specifications or design mix.

#### **5. Code:**

- IS 2386 (Part 1): 1963 – Methods of Test for Aggregates for Concrete – Part 1: Particle Size and Shape.
- IS 383:2016 – Coarse and Fine Aggregate for Concrete – Specification.

#### **➤ Flakiness Index Test:**

##### **1. Aim of the Test:**

- To determine the percentage of flaky particles in coarse aggregates and assess their suitability for use in pavement and concrete works.

##### **2. Theory:**

- Flaky particles are those whose least dimension (thickness) is less than 0.6 times the mean size of the aggregate. Excess flaky particles reduce the strength, interlocking, and durability of pavement layers. The test determines the percentage by weight of flaky aggregates in a given sample.

##### **3. Procedure (As per IS:2386 Part 1:1963):**

- Separate the aggregate sample into different size fractions using standard IS sieves.
- Weigh each size fraction separately.
- Pass each particle through the corresponding slot of the flakiness gauge.
- Particles passing through the slot are considered flaky.
- Weigh the flaky particles from each size fraction.
- Calculate the Flakiness Index as the percentage of the total weight of flaky particles to the total weight of the sample.

##### **4. Acceptance / Standards (As per IS:383-2016):**

- Flakiness Index should not exceed 30% for aggregates used in concrete and pavement works.

- Where both flakiness and elongation are determined, the combined Flakiness and Elongation Index should not exceed 35%.

5. **Code:**

- IS:2386 (Part 1):1963 – Methods of Test for Aggregates for Concrete – Part 1: Particle Size and Shape
- IS:383:2016 – Coarse and Fine Aggregate for Concrete – Specification

➤ **Elongation Index Test:**

1. **Aim of the Test:**

- To determine the percentage of elongated particles in coarse aggregates and assess their suitability for use in pavement and concrete works.

2. **Theory:**

- Elongated particles are those whose greatest dimension (length) is greater than 1.8 times the mean size of the aggregate. A high percentage of elongated particles reduces the strength, workability, and interlocking of aggregates, thereby affecting the performance of pavement and concrete.

3. **Procedure (As per IS:2386 Part 1:1963):**

- Separate the aggregate sample into different size fractions using standard IS sieves.
- Weigh each size fraction separately.
- Pass each particle through the corresponding opening of the elongation gauge.
- Particles retained on the gauge are considered elongated.
- Weigh the elongated particles from each size fraction.
- Calculate the Elongation Index as the percentage of the total weight of elongated particles to the total weight of the sample.

4. **Acceptance / Standards (As per IS:383-2016):**

- No separate limit is specified for Elongation Index in IS:383:2016.
- Where both flakiness and elongation are determined, the combined Flakiness and Elongation Index shall not exceed 35% for aggregates used in concrete and pavement works.

5. **Code:**

- IS:2386 (Part 1):1963 – Methods of Test for Aggregates for Concrete – Part 1: Particle Size and Shape
- IS:383:2016 – Coarse and Fine Aggregate for Concrete – Specification

➤ **Aggregate Impact Value (AIV) Test:**

1. **Aim of the Test:**

- To determine the toughness of coarse aggregates and their resistance to sudden impact loads for use in road and pavement construction.

2. **Theory:**

- The Aggregate Impact Value (AIV) test measures the percentage of fines produced when aggregates are subjected to a specified number of blows under standard impact conditions. Lower AIV indicates tougher aggregates with higher resistance to impact.

3. **Procedure (As per IS:2386 Part 4:1963):**

- Take aggregates passing the 12.5 mm IS sieve and retained on the 10 mm IS sieve.
- Fill the impact testing mould in three equal layers, tamping each layer with 25 strokes of the tamping rod.
- Level the surface and fix the mould in the impact testing machine.
- Subject the sample to 15 blows from a 13.5–14.0 kg hammer falling freely from a height of 380 mm.
- Remove the sample and sieve it through the 2.36 mm IS sieve.
- Weigh the material passing the 2.36 mm sieve and calculate the Aggregate Impact Value as the percentage of fines to the total weight of the sample.

4. **Acceptance / Standards (As per IS:383-2016):**

- Aggregate Impact Value  $\leq 30\%$  for aggregates used in wearing surfaces.
- Aggregate Impact Value  $\leq 45\%$  for aggregates used in base and sub-base courses.

5. **Code:**

- IS:2386 (Part 4):1963 – Methods of Test for Aggregates for Concrete – Part 4: Mechanical Properties
- IS:383:2016 – Coarse and Fine Aggregate for Concrete – Specification

➤ **Aggregate Crushing Value (ACV) Test:**

**1. Aim of the Test:**

- To determine the resistance of coarse aggregates to crushing under a gradually applied compressive load and assess their suitability for road and pavement construction.

**2. Theory:**

- The Aggregate Crushing Value (ACV) test measures the percentage of fines produced when a standard aggregate sample is subjected to a specified compressive load. A lower ACV indicates stronger aggregates with higher resistance to crushing.

**3. Procedure (As per IS:2386 Part 4:1963):**

- Take aggregate passing the 12.5 mm IS sieve and retained on the 10 mm IS sieve.
- Fill the cylindrical measure with the aggregate in three equal layers, tamping each layer with 25 strokes of the tamping rod.
- Transfer the sample to the compression mould and level the surface.
- Place the mould in the compression testing machine and apply a load of 400 kN gradually over 10 minutes.
- Remove the crushed aggregate and sieve it through the 2.36 mm IS sieve.
- Weigh the material passing the 2.36 mm sieve and calculate the Aggregate Crushing Value as the percentage of fines to the total weight of the sample.

**4. Acceptance / Standards (As per IS:383-2016):**

- Aggregate Crushing Value  $\leq$  30% for aggregates used in wearing surfaces.
- Aggregate Crushing Value  $\leq$  45% for aggregates used in base and sub-base courses.

**5. Code:**

- IS:2386 (Part 4):1963 – Methods of Test for Aggregates for Concrete – Part 4: Mechanical Properties
- IS:383:2016 – Coarse and Fine Aggregate for Concrete – Specification

## **CONCRETE TESTING:**

To determine the workability, strength, and quality of concrete, ensuring it meets the specified design and performance requirements for construction. Concrete testing helps verify properties such as slump, compressive strength, and overall quality, thereby ensuring proper placement, adequate strength development, durability, and structural stability while preventing defects, maintaining quality control, and enhancing the safety, service life, and cost-effectiveness of the project.

## **TEST FOR CONCRETE:**

### ➤ **Compressive Strength of Concrete Test:**

#### **1. Aim of the Test:**

- To determine the compressive strength of hardened concrete and verify its suitability for structural and pavement works.

#### **2. Theory:**

- Compressive strength is the maximum compressive stress that concrete can withstand before failure. It is determined by applying a gradually increasing compressive load on standard concrete cubes using a Compression Testing Machine (CTM). The test indicates the quality and load-bearing capacity of concrete.

#### **3. Procedure (As per IS:516-1959):**

- Cast concrete cubes of 150 mm × 150 mm × 150 mm and compact them properly.
- Cure the specimens in clean water for the specified period (generally 7 and 28 days).
- Remove the cubes from water and wipe off excess moisture.
- Place the cube centrally in the Compression Testing Machine (CTM).
- Apply the load uniformly at a rate of approximately 140 kg/cm<sup>2</sup>/min (14 N/mm<sup>2</sup>/min) until the specimen fails.
- Record the maximum load at failure and calculate the compressive strength by dividing the maximum load by the loaded area of the cube.

#### **4. Acceptance / Standards (As per IS:456-2000):**

- The 28-day compressive strength shall be not less than the characteristic compressive strength (f<sub>ck</sub>) of the specified grade of concrete.
- Examples:  
M20 Concrete: ≥ 20 MPa  
M25 Concrete: ≥ 25 MPa

M30 Concrete:  $\geq 30$  MPa

**5. Code:**

- IS:516-1959 – Method of Tests for Strength of Concrete
- IS:456-2000 – Plain and Reinforced Concrete – Code of Practice

➤ **Workability of Concrete by Slump Cone Test:**

**1. Aim of the Test:**

- To determine the workability and consistency of fresh concrete using the slump cone method.

**2. Theory:**

- Workability is the property of fresh concrete that determines the ease of mixing, transporting, placing, compacting, and finishing without segregation or bleeding. The slump test measures the vertical settlement (slump) of fresh concrete after removal of the mould, indicating its consistency and workability.

**3. Procedure (As per IS:516-1959):**

- Clean and moisten the slump cone and place it on a rigid, non-absorbent surface.
- Fill the cone with fresh concrete in four equal layers.
- Compact each layer with 25 strokes using a standard tamping rod.
- Strike off the excess concrete to level the top surface.
- Lift the cone vertically upward slowly in 5–10 seconds without disturbing the concrete.
- Measure the vertical difference between the height of the mould and the highest point of the slumped concrete. This value is reported as the slump in millimetres.

**4. Acceptance / Standards (As per IS:456-2000):**

As per IS:456:2000 (Recommended Slump Values):

- Mass concrete: 25–50 mm
- RCC beams and slabs: 25–75 mm
- Columns and thin sections: 75–100 mm
- Pumped concrete: 75–150 mm

As per Approved Design Mix / Project Specifications:

- The measured slump shall be within the specified slump range of the approved design mix (e.g., PQC generally 20–40 mm, unless otherwise specified).

**5. Code:**

- IS:1199 – Methods of Sampling and Analysis of Concrete – Determination of Workability of Fresh Concrete.
- IS:456:2000 – Plain and Reinforced Concrete – Code of Practice.

**Cemindia**  
ANNEXURE-5/13

Contract No: EPC 2489-2024  
Metro Railway Kolkata

Source of Material: Panchami (Nahati)  
Size of Material: 20 mm  
Sampled By: Cemindia Projects Limited & Metro Railway

Aggregate Crushing Value As per IS:2386 Part-IV  
QR-11

Date of Sampling: 25-03-2026  
Date of Testing: 25-03-2026

| Determination                                                                | 1     | 2     |
|------------------------------------------------------------------------------|-------|-------|
| Weight of Sample Passing 15 Sieve 12.5 mm and retained on 15 Sieve 10mm (gm) | 350   | 350   |
| Weight of Fraction Retained on 15 Sieve 2.36 mm after the Impact Test (gm)   | 213   | 187   |
| Weight of Fraction Passing 15 Sieve 2.36 mm after the Impact Test (gm)       | 137   | 163   |
| Impact Value of Aggregate (%)                                                | 14.49 | 13.45 |

Remarks: OK

Signature: Amab Kar  
Date: 24/06/26

**Cemindia**  
ANNEXURE-5/13

Contract No: EPC 2489-2024  
Metro Railway Kolkata

Source of Material: Panchami (Nahati)  
Size of Material: 20 mm  
Sampled By: Cemindia Projects Limited & Metro Railway

Aggregate Crushing Value As per IS:2386 Part-IV  
QR-11

Date of Sampling: 27-06-2026  
Date of Testing: 27-06-2026

| Determination                                                                | 1     | 2     |
|------------------------------------------------------------------------------|-------|-------|
| Weight of Sample Passing 15 Sieve 12.5 mm and retained on 15 Sieve 10mm (gm) | 354   | 334   |
| Weight of Fraction Retained on 15 Sieve 2.36 mm after the Impact Test (gm)   | 309   | 291   |
| Weight of Fraction Passing 15 Sieve 2.36 mm after the Impact Test (gm)       | 45    | 43    |
| Impact Value of Aggregate (%)                                                | 12.71 | 12.87 |

Remarks: OK

Signature: Amab Kar  
Date: 24/06/26

**Cemindia**  
ANNEXURE-5/13

Contract No: EPC 2489-2024  
Metro Railway Kolkata

Source of Material: Panchami (Nahati)  
Size of Material: 10 mm  
Sampled By: Cemindia Projects Limited & Metro Railway

Aggregate Crushing Value As per IS:2386 Part-IV  
QR-11

Date of Sampling: 27-06-2026  
Date of Testing: 27-06-2026

| Determination                                                                | 1     | 2     |
|------------------------------------------------------------------------------|-------|-------|
| Weight of Sample Passing 15 Sieve 12.5 mm and retained on 15 Sieve 10mm (gm) | 352   | 357   |
| Weight of Fraction Retained on 15 Sieve 2.36 mm after the Impact Test (gm)   | 301   | 309   |
| Weight of Fraction Passing 15 Sieve 2.36 mm after the Impact Test (gm)       | 51    | 48    |
| Impact Value of Aggregate (%)                                                | 14.49 | 13.45 |

Remarks: OK

Signature: Amab Kar  
Date: 24/06/26

**Cemindia**  
ANNEXURE-5/24

Contract No: EPC 2489-2024  
Metro Railway Kolkata

Source of Material: Panchami (Nahati)  
Size of Material: 20 mm  
Sampled By: Cemindia Projects Limited & Metro Railway

Aggregate Crushing Value As per IS:2386 Part-IV  
QR-24

Date of Sampling: 27-06-2026  
Date of Testing: 27-06-2026

| Sl. No. | Determination                                                                                    | Test 1 | Test 2 | Average |
|---------|--------------------------------------------------------------------------------------------------|--------|--------|---------|
| 1       | Weight of Mould W <sub>1</sub> (gm)                                                              | 3068   | 3068   |         |
| 2       | Weight of Mould + Sample W <sub>2</sub> (gm)                                                     | 5812   | 5846   |         |
| 3       | Weight of Sample W <sub>3</sub> (gm)                                                             | 2744   | 2778   |         |
| 4       | Weight of sample retained over 2.36 mm sieve after crushing (W <sub>4</sub> ) (gm)               | 2242   | 2267   |         |
| 5       | Weight of sample passing over 2.36 mm sieve after crushing (W <sub>5</sub> ) (gm)                | 499    | 506    |         |
| 6       | Loss in Wt. of aggregate W <sub>6</sub> = W <sub>2</sub> - (W <sub>4</sub> + W <sub>5</sub> ) gm | 3      | 5      |         |
| 7       | Aggregate Crushing value = W <sub>6</sub> / 100 / W <sub>3</sub> (%)                             | 18.19  | 18.21  | 18.20   |

Remarks: OK

Signature: Amab Kar  
Date: 24/06/26

Some pictures of Test Result

## CONCLUSION

### Internship Exposure

- Diaphragm wall construction and reinforcement inspection
- Excavation and earth-retaining systems
- Concrete quality control and testing
- Surveying and site measurements
- QA/QC documentation and inspection procedures
- Daily progress monitoring and reporting
- Safety practices and toolbox meetings
- Coordination among civil, structural and MEP teams

The industrial training at Cemindia Projects Limited on the Birati Underground Metro Project, Metro Railway, Kolkata, provided me with valuable practical exposure to the execution of underground metro infrastructure. During the training, I gained a clear understanding of modern construction techniques, engineering practices, and the importance of safety and quality in large-scale civil engineering projects.

I learned about diaphragm wall and guide wall construction, sheet piling, reinforcement works, concrete batching plant operations, material testing, quality control, and various activities involved in underground station construction. This experience helped me bridge the gap between theoretical knowledge and its practical application on site.

The training also allowed me to observe the professional responsibilities of engineers in planning, supervision, and execution, while emphasizing the importance of teamwork, coordination, and adherence to engineering standards.

I sincerely thank the engineers and staff of Cemindia Projects Limited for their guidance and support throughout the training. Their mentorship made the experience both informative and rewarding.

Overall, this industrial training significantly enhanced my technical knowledge and practical skills, providing a strong foundation for my academic development and future career as a civil engineer.